Basic Principles of Medicine 1
Module : Foundations to Medicine
Lecture No: 48

Title: Genetics of Tumor Suppressor Genes

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Recommended Reading

- **Human Genetics and Genomics, Korf and Irons, 4th Edition Chapter 8 and 15**
  - **You can find the following pages and chapters on your Sakia site under:**
  - Resources/General Information/Book Access

Genetics of Tumor Suppressor Genes - Objectives

SOM.MK.III.BPM1.1.FTM.3.GNET.0139	Distinguish the role of tumor suppressor genes including: mutator genes (DNA repair; caretakers); genes involved in cell cycle regulation (gatekeepers); and genes involved in apoptosis regulation
SOM.MK.III.BPM1.1.FTM.3.GNET.0140	Indicate the genetic basis of Li-Fraumeni syndrome
SOM.MK.III.BPM1.1.FTM.3.GNET.0141	Analyze the two-hit model where loss of heterozygosity for the action of tumor suppressor genes (inherited mutation vs. sporadic) to understand how the inheritance pattern differs between sporadic and inherited Rb
SOM.MK.III.BPM1.1.FTM.3.GNET.0142	Interpret familial cancer syndromes and their genetic principles to describe how a predisposition to cancer can be transmitted to offspring use BRCA and APC as examples.
SOM.MK.III.BPM1.1.FTM.3.GNET.0143	Determine the characteristics of inherited colorectal (familial adenomatous polyposis and hereditary nonpolyposis colon cancer) and breast cancers
SOM.MK.III.BPM1.1.FTM.3.GNET.0144	Differentiate the clinical presentation of inherited cancers from that of sporadic cancers
SOM.MK.III.BPM1.1.FTM.3.GNET.0145	Indicate the utility of expression array analysis in cancer
SOM.MK.III.BPM1.1.FTM.3.GNET.0146	Identify the role of epigenetic changes and microRNA in tumorigenesis

Required Reading and Objectives

After 3 lectures on Cancer Genetics: Genetics of Oncogenes; Genetics of Tumor Suppressor Genes & DNA Repair.

Apoptosis and Cancer

1. Lippincott Illustrated Reviews: Cell and Molecular Biology, Chapter 22

[Abnormal Cell Growth | Lippincott® Illustrated Reviews: Cell and Molecular Biology, 3e | Premium Basic Sciences | Health Library (lwwhealthlibrary.com)]

The following sections only:
Genes and Cancer: Abnormal cell growth: p53 – the guardian of the genome

2. Lippincott Illustrated Reviews: Cell and Molecular Biology, Chapter 23

[Cell Death | Lippincott® Illustrated Reviews: Cell and Molecular Biology, 3e | Premium Basic Sciences | Health Library (lwwhealthlibrary.com)]

The following sections only:
Cell Death: Initiation of apoptosis Including figure 23.7
Bcl-2 family including figures 23.13 and 23.14
Apoptosis in disease
Cancer

SOM.MK.III.BPM1.1.FTM.3.GNET.0150	Discuss the central role of p53 in regulation of cell proliferation, cell survival, and apoptosis
SOM.MK.III.BPM1.1.FTM.3.GNET.0151	Discuss the role of BAX and BCL-2 in regulating apoptosis: -BAX as pro-apoptotic, -BCL-2 as anti-apoptotic
SOM.MK.III.BPM1.1.FTM.3.GNET.0152	Indicate how cytochrome c release from mitochondria may initiate apoptosis
SOM.MK.III.BPM1.1.FTM.3.GNET.0153	Summarize the extrinsic, intrinsic, and perforin pathway of apoptosis with respect to cancer.

Cancer Causing Genes

1. **Oncogenes** – normally stimulate growth
2. **Tumor Suppressor genes –** which normally inhibit growth
3. **DNA Repair genes** - normally limit mutations

Many inherited predispositions to cancer result from mutations in tumor suppressor genes

Inherited Autosomal Dominant Cancer Syndromes (recessive at the cellular level, but with guaranteed loss of second copy)

Inherited Autosomal Recessive Cancer Syndromes of Defective DNA Repair
Xeroderma pigmentosum; Ataxia-telangiectasia; Bloom syndrome; Fanconi anemia

Associated with a single, susceptibility gene

Familial Cancers
Familial clustering, but role of inherited predisposition not clear for each individual: Breast cancer; Ovarian cancer; Pancreatic cancer

Multifactorial cause

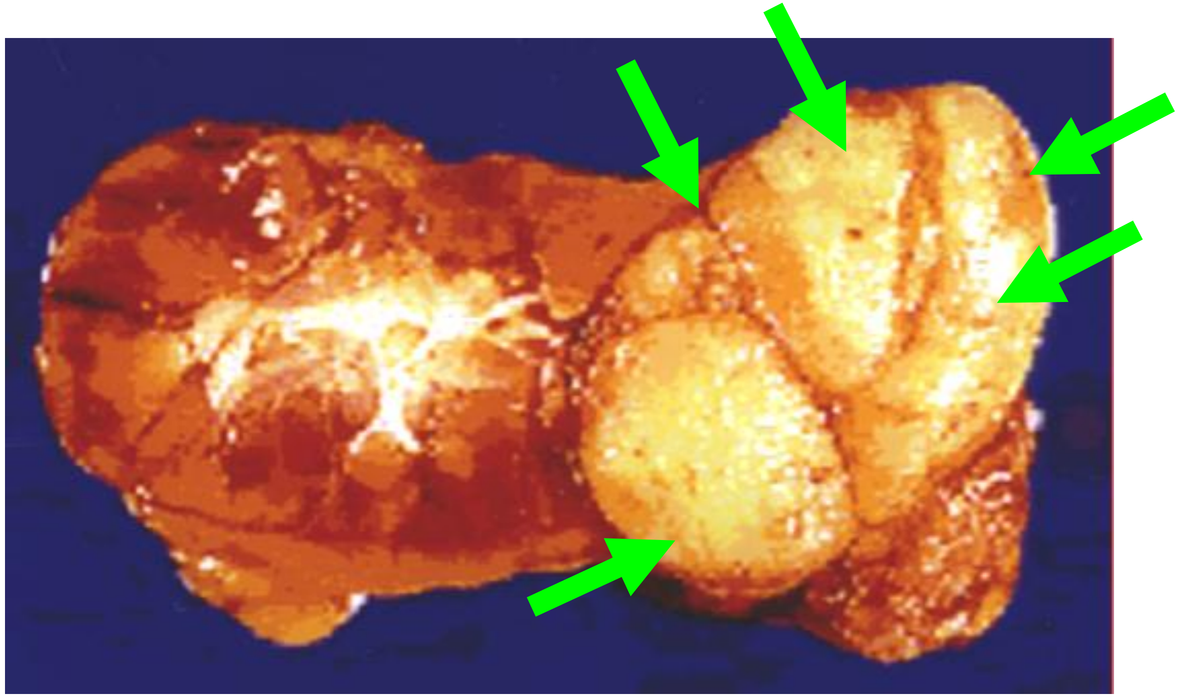
Tumor Suppressor Genes

(Korf, Table 8.1, modified)

SOM.MKIII.BPM1.1.FTM.3.GNET.0139

GENE	SYNDROME	MAJOR TUMORS
RB	Retinoblastoma	Retinoblastoma, osteosarcoma
TP53	Li-Fraumeni Syndrome	Soft-tissue sarcoma, glioma, leukemia
APC	Familial adenomatous polyposis (FAP)	Colorectal cancer
MSH2, MLH1, PMS1/2	Hereditary nonpolyposis colon cancer (HNPCC)	Colorectal cancer
BRCA1/2	Familial breast/ovarian cancer	Breast and ovarian cancer
VHL	Von Hippel-Lindau Syndrome	Hemangioblastoma, pheochromocytoma, renal cell carcinoma
NF1	**Neurofibromatosis 1**	Neurofibroma, astrocytoma, sarcoma, leukemia
NF2	Neurofibromatosis 2	Schwannoma, meningioma, ependymoma
TSC1/2	Tuberous sclerosis complex	Cortical dysplasia, renal angiomyolipoma
WT1	WAGR (Wilms tumor, aniridia, genitourinary anomalies, growth retardation)	Wilms tumor
MEN1	Multiple endocrine neoplasia 1	Islet cell adenoma, pituitary adenoma, parathyroid adenoma

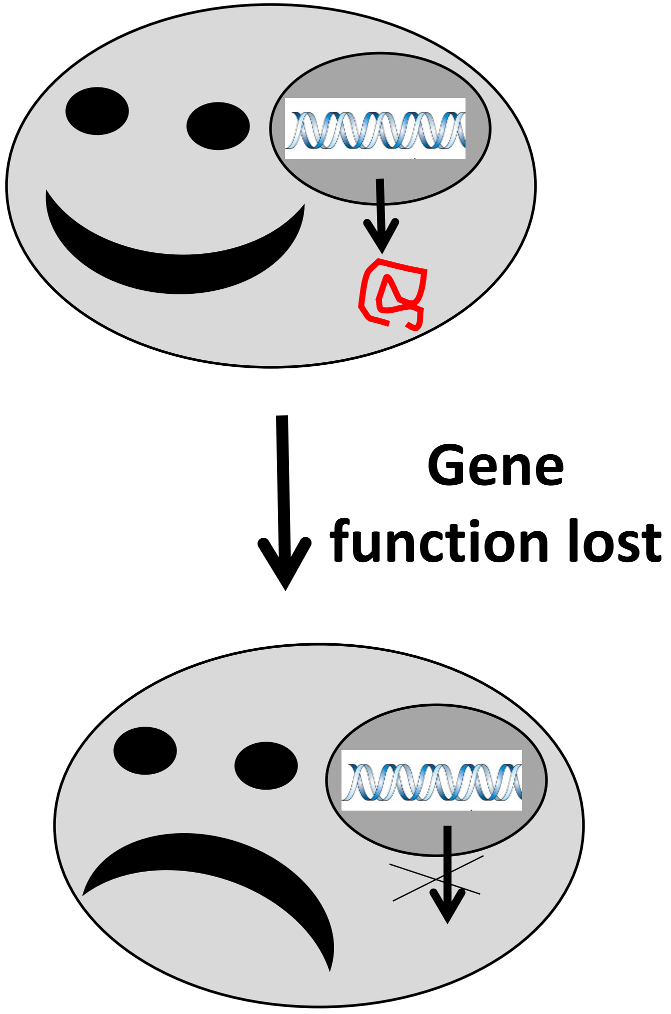
Wilms Tumor: Autosomal Dominant Inheritance



Results from **loss-of-function** in the **WT1** gene on Chromosome 11, which **encodes a transcription factor** important in the control of cell growth and differentiation. (Renal cancer)

Tumor Suppressor Genes

- Genes that cause cancer when they lose their function ('**Loss of function**' mutation)
- The normal function of these proteins is to stop cancer (**control growth**)
- They are cell cycle control genes, apoptosis promoting genes and/or DNA repair genes.
- These genes code for proteins that control cell cycle, proteins that promote apoptosis and DNA repair genes



Therefore, one 'good' gene is enough for cell cycle control, you need to lose the function of both to get cancer = "two hits"

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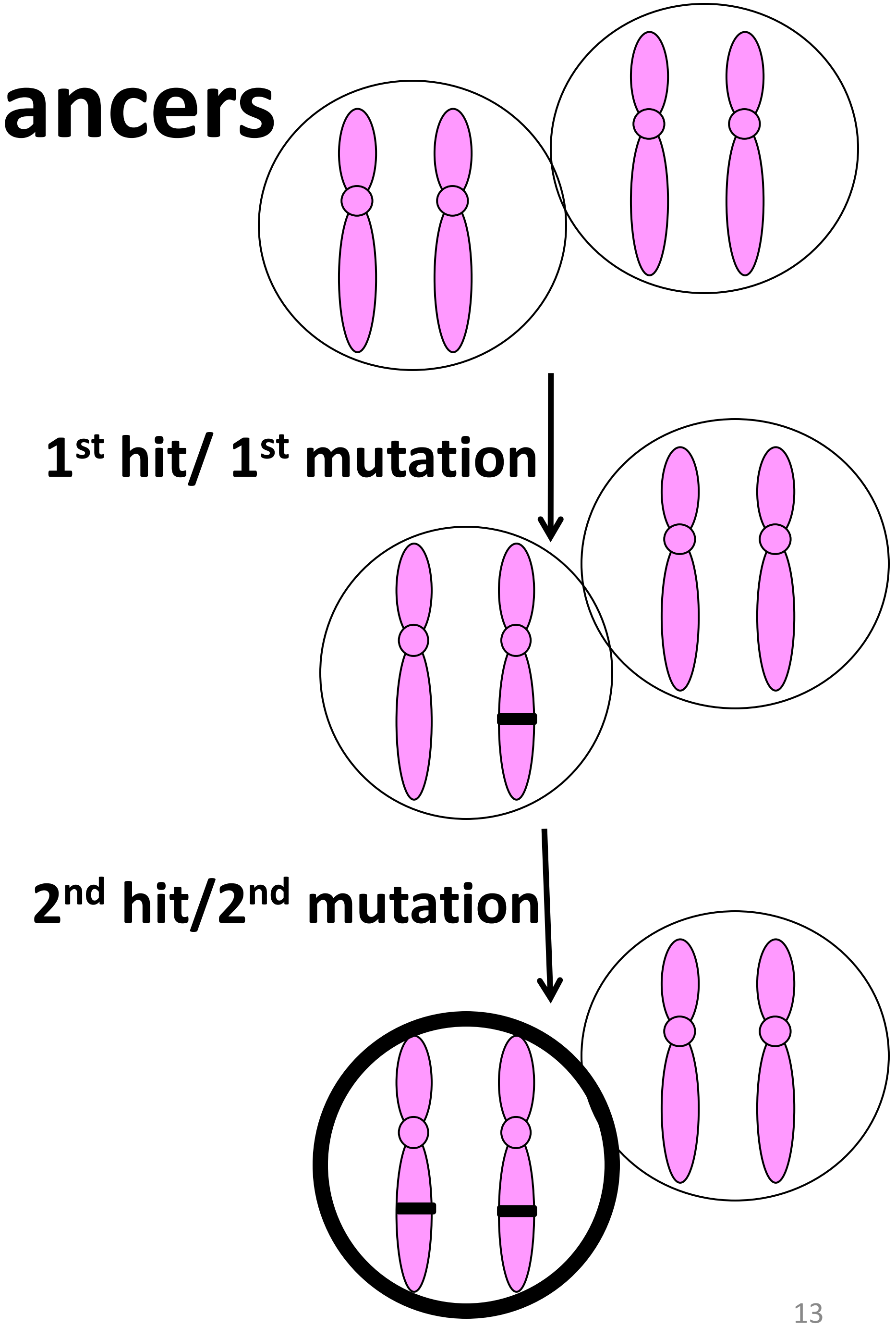
Two Hit Hypothesis: Sporadic cancers

- A mutation occurs in a tumor suppressor gene on one homolog in one cell (1^{st} hit)
- The second mutation occurs in the same gene (2^{nd} copy) in the same cell
- Loss of all tumor suppressor activity
- Cancer/ malignant transformation



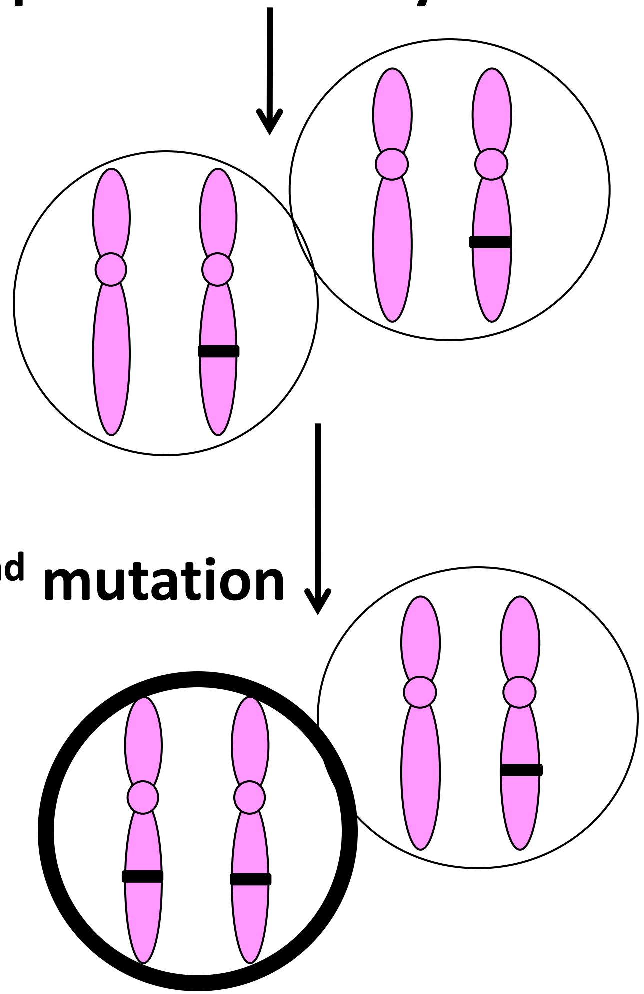
'Two hits'/ two mutations are needed for tumor initiation

Two Hit Hypothesis: Familial cancers

- First mutation is inherited, and therefore, present in every cell (**1st hit**)
- Second mutation can occur in <u>any</u> cell (**2nd hit**)
- Loss of all tumor suppressor activity in that cell
- Cancer/ malignant transformation

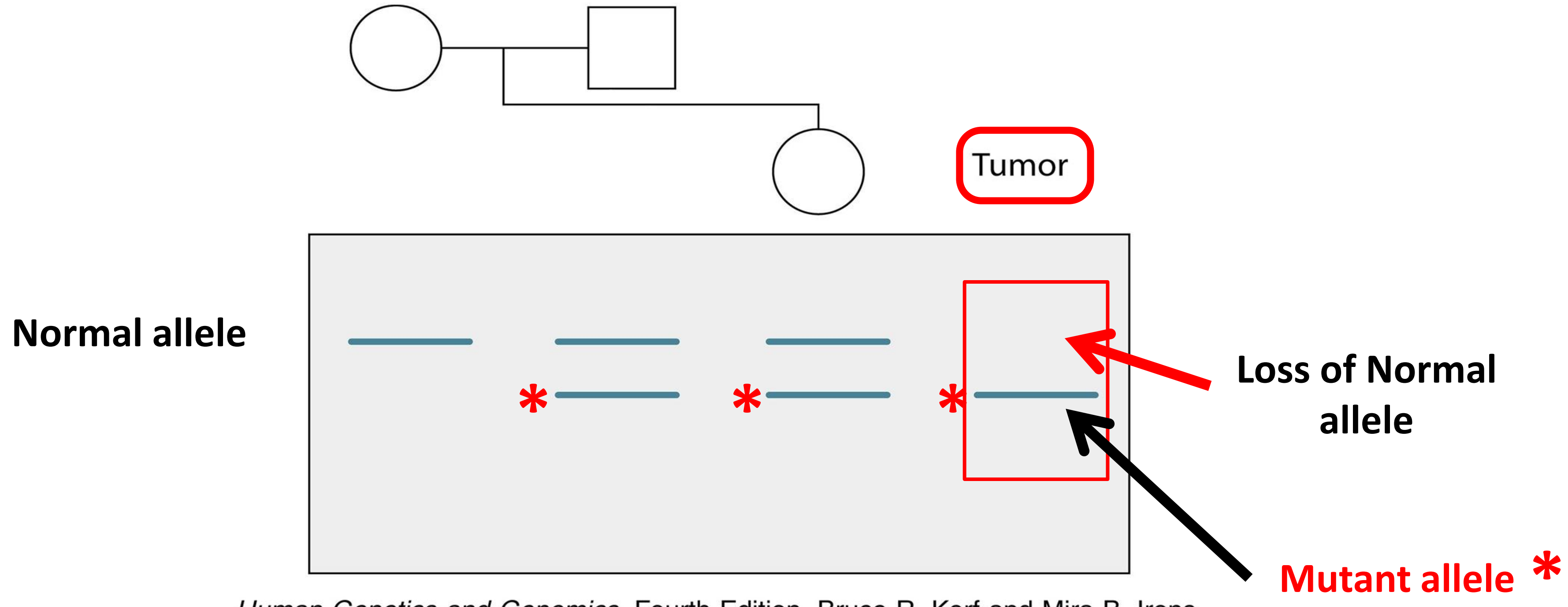
The 1st hit is present in every cell (inherited). Only 'one more hit' (2nd hit) needed for tumor initiation

1st hit/ 1st mutation (inherited) is present in every cell



2nd hit/2nd mutation

Second 'Hit' usually results in **Loss of Heterozygosity (LOH)**



Human Genetics and Genomics, Fourth Edition. Bruce R. Korf and Mira B. Irons.

Figure 8.8 Loss of heterozygosity for polymorphic marker on chromosome 13. DNA from the mother, father, and daughter, as well as the daughter's retinoblastoma tumor, was analyzed for a polymorphism involving DNA sequence from chromosome 13q14, demonstrating two alleles of different size (producing an upper and lower band after gel electrophoresis). The mother is homozygous for the upper allele, and the father and affected daughter are heterozygous. DNA from the retinoblastoma, however, shows only the lower allele, indicating loss of the upper allele in the tumor.

Explained on next slide

Second 'Hit' usually results in **Loss of Heterozygosity (LOH)**

- Refer to figure in the previous slide
- Mom is homozygous for the normal allele of Rb gene
- Dad and child are heterozygous (Contain a mutant allele and a normal allele)
- But the Rb tumor tissue has only a single allele (mutant allele; The normal allele is lost)
- In the tumor tissue, there is loss of the normal allele
- Tumor tissue is homozygous for the mutant allele
- LOH (**Loss of heterozygosity**) for the Rb gene in the tumor (Two hit hypothesis)

Mechanisms producing 2nd hit

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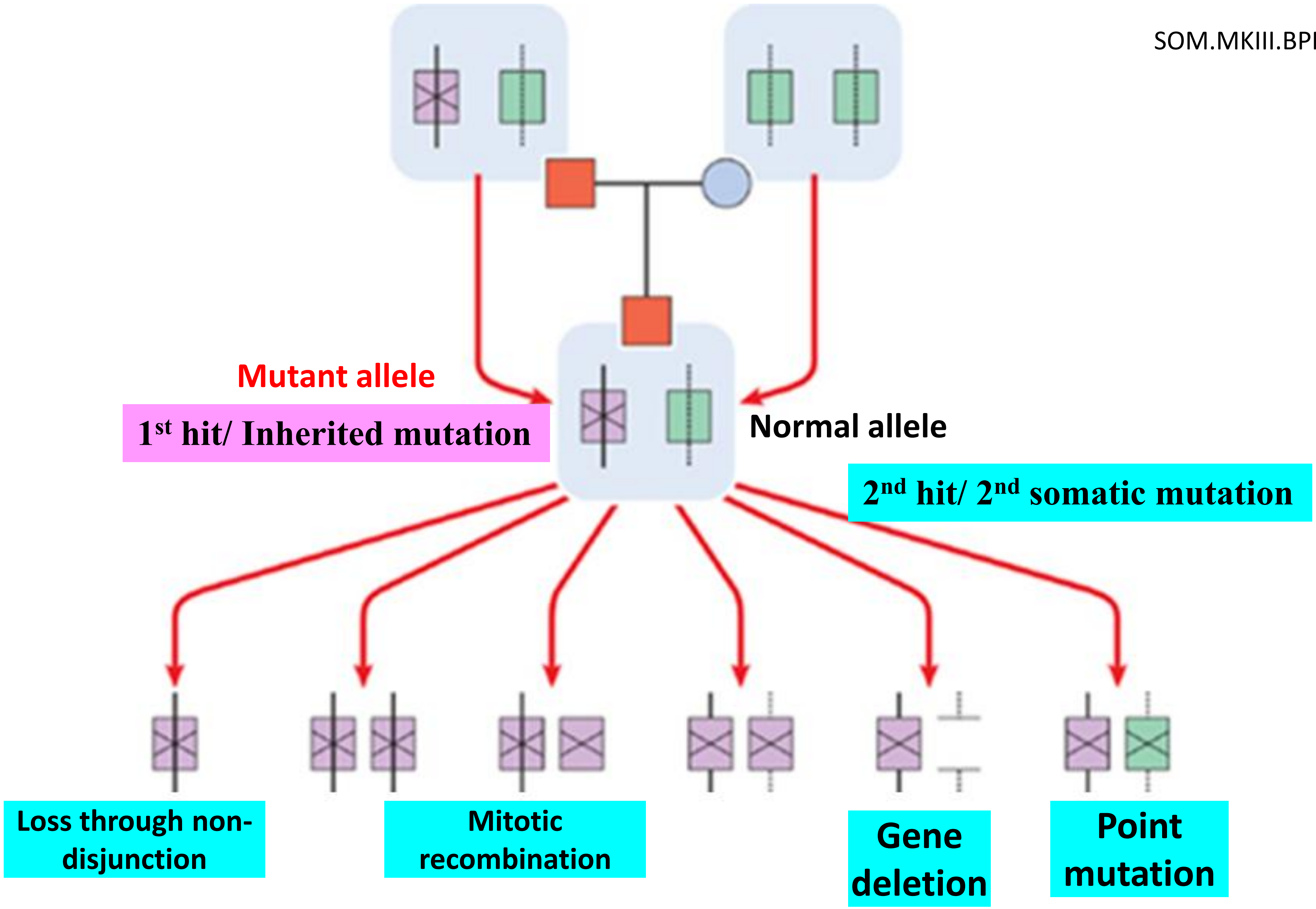
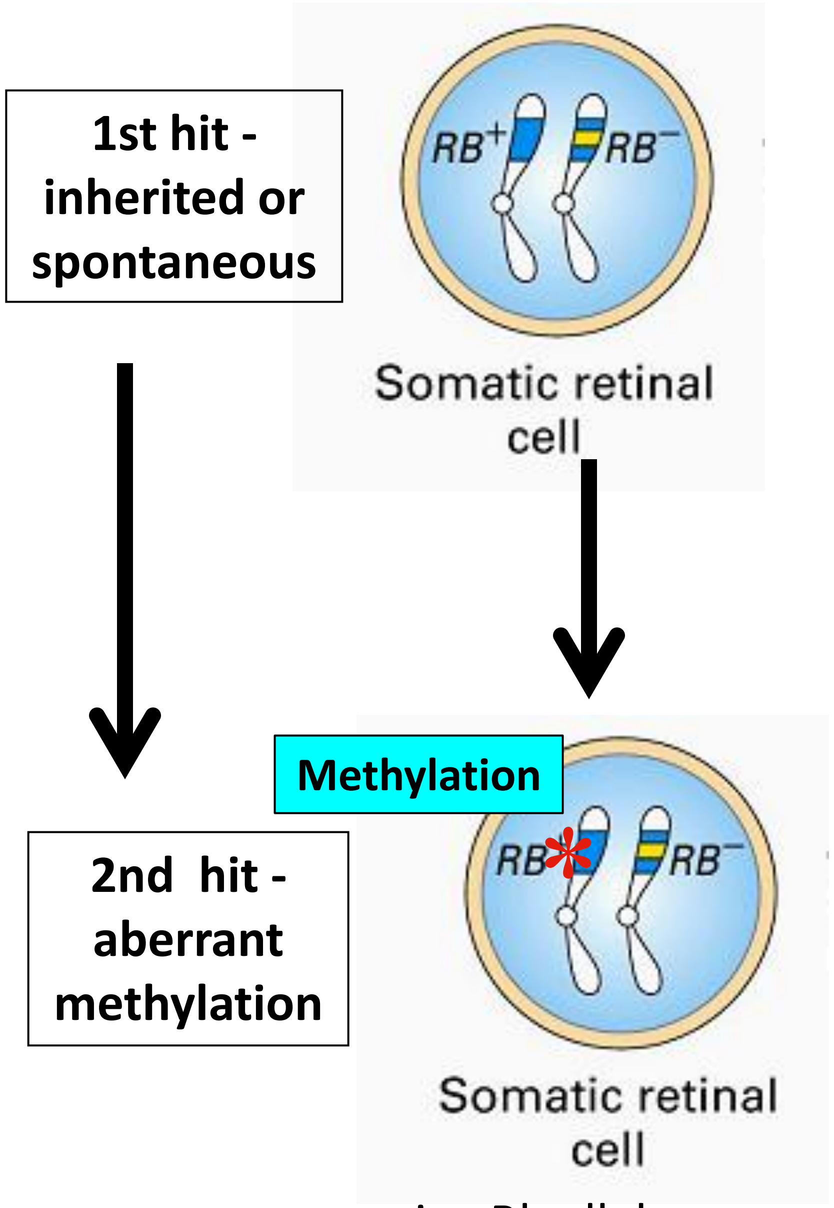


Mechanisms producing 2nd hit

- Sometimes aberrant methylation of the gene results in gene silencing ***(Epigenetic mechanism)***
- The gene is still present but mistakenly silenced (methylated)
- Mutations in DNA silencing mechanisms can also result in tumors



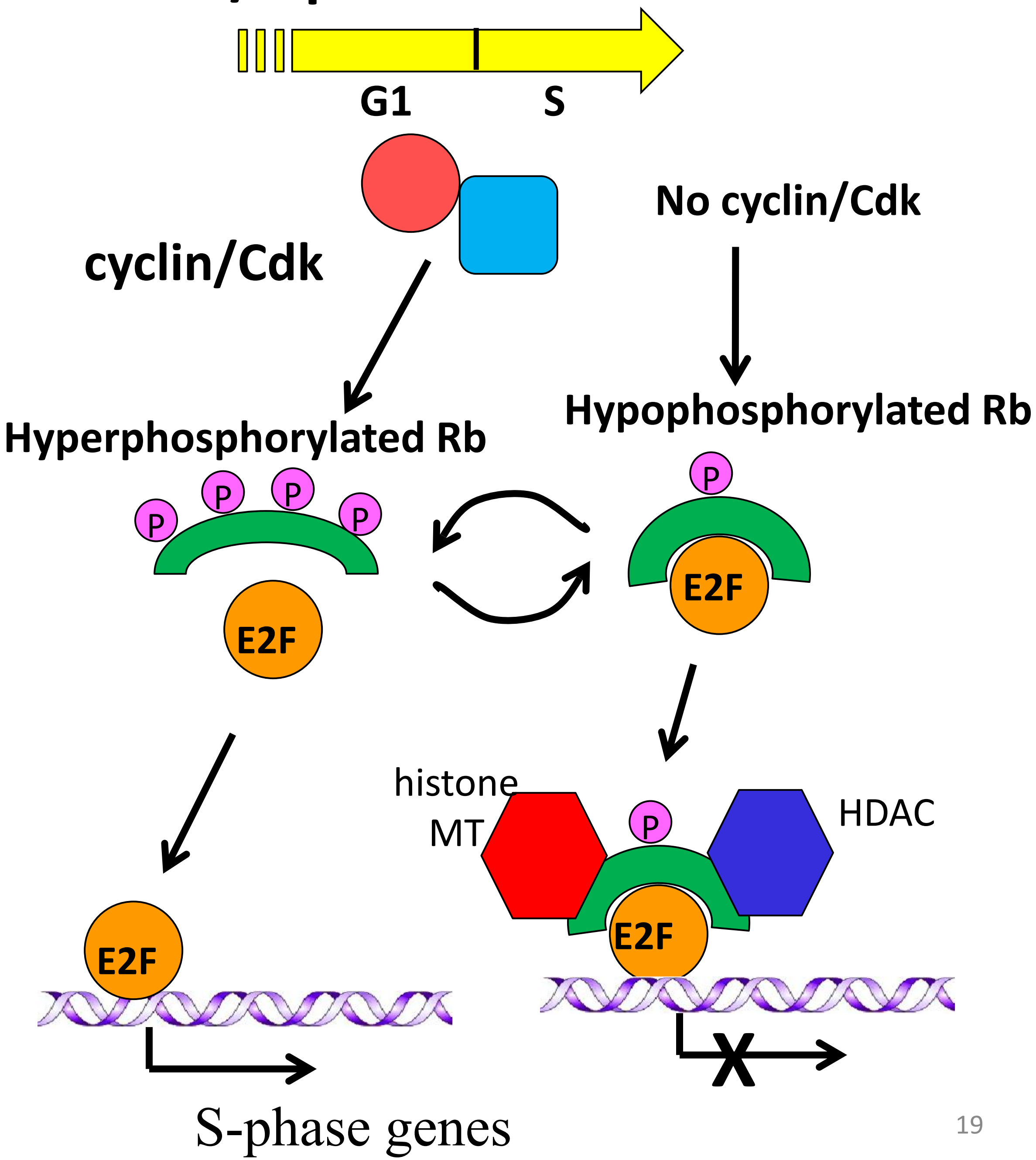
no active Rb alleles = tumor

Rb protein – Regulator of G1/S phase transition

At G1/S checkpoint

- Cyclin/Cdks hyperphosphorylate Rb
- Rb no longer represses E2F
- E2F activates S-phase genes (Cell divides)

- **No cyclin/Cdk**
- Rb/E2F complex still binds DNA
- Recruits histone methylase and histone deacetylase
- Result in transcriptional block and G1 arrest



Rb protein – Regulator of G1/S phase transition

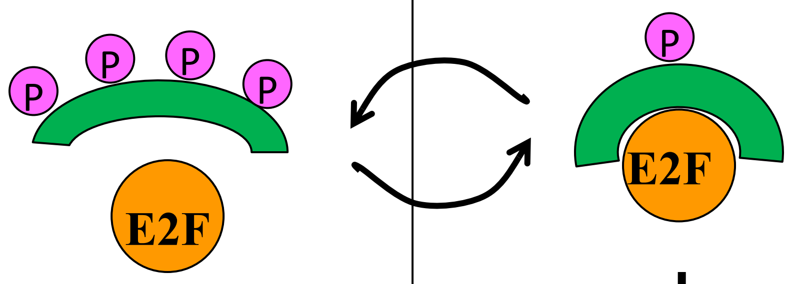
Inactivation of Rb is the critical final step in driving G1-S transition (i.e. cell division)

Loss of Rb/ mutant Rb, does NOT bind to E2F → Increased transcription of S phase genes (Unregulated cell division)

Mutant Rb/ No Rb protein

Hyperphosphorylated Rb

Hypophosphorylated Rb



Cell Division

No G1 arrest (cell cycle progresses)

NO Cell Division

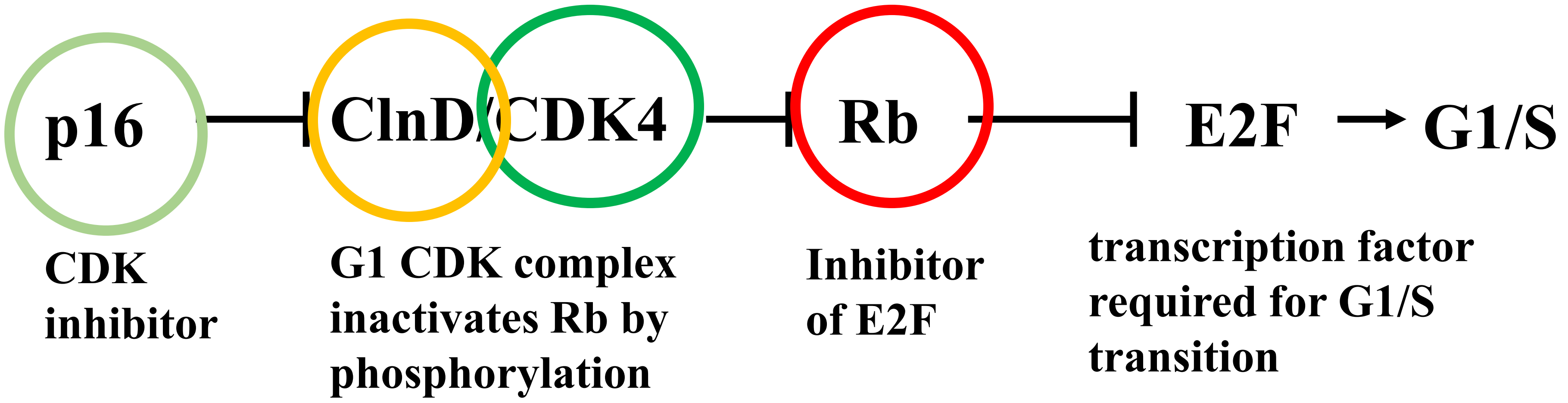
↓

G1 arrest (cell cycle arrest)

Rb protein – Regulator of G1/S phase transition

Virtually all cancer cells show **dysregulation of the G_1-S checkpoint** as a result of mutation in one of four genes that regulate the phosphorylation of RB

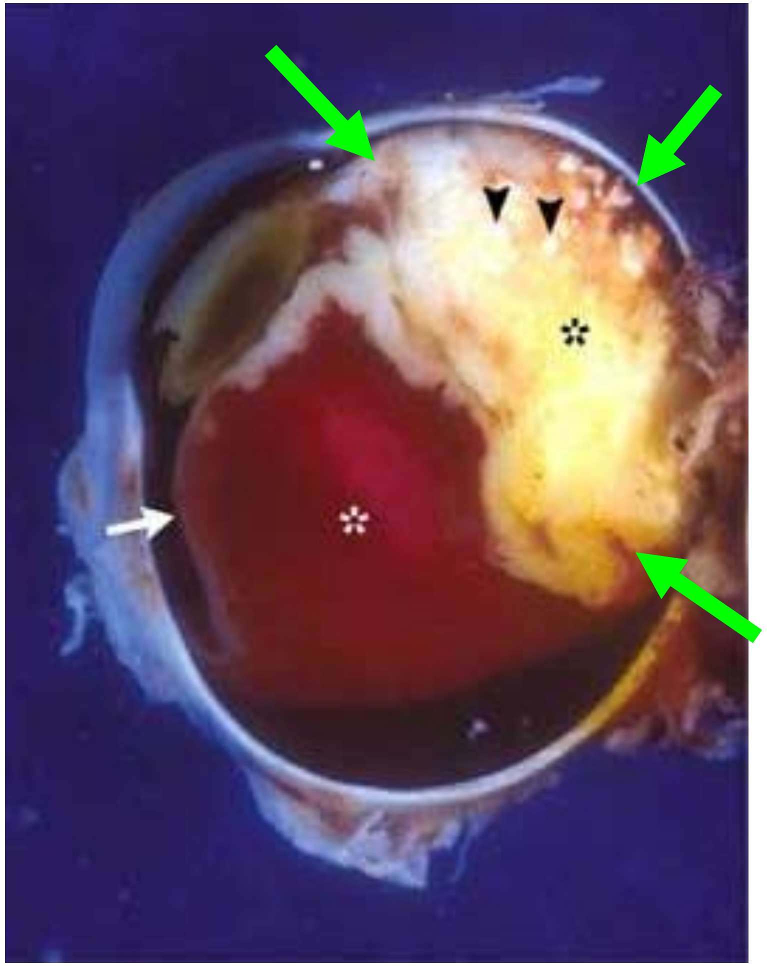
The four genes are *RB*, *CDK4*, cyclin D gene, and *CDKN2A* (p16).



Retinoblastoma (a childhood cancer)

https://www.cancer.org/cancer/retinoblastoma.html

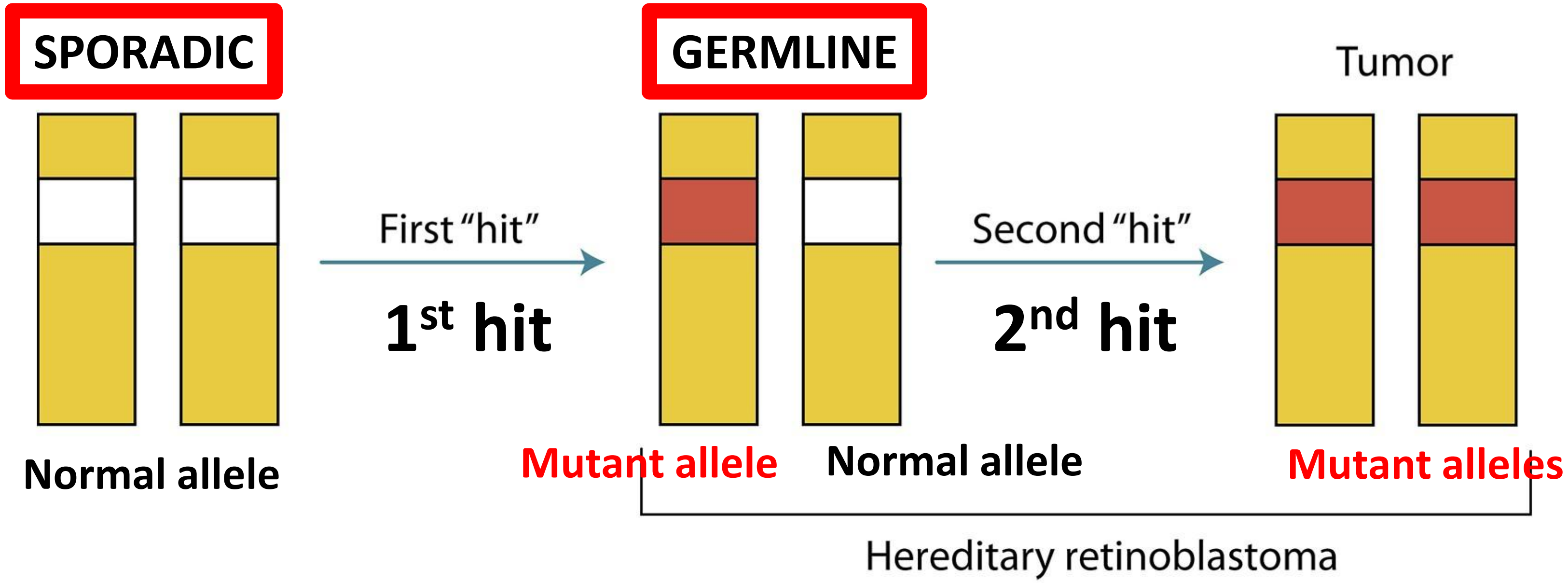
'White pupillary reflex' in the eye



Incidence of about 1 in 20,000 births

Mutation of Rb gene on chromosome 13

Familial and non-familial (sporadic) Retinoblastoma



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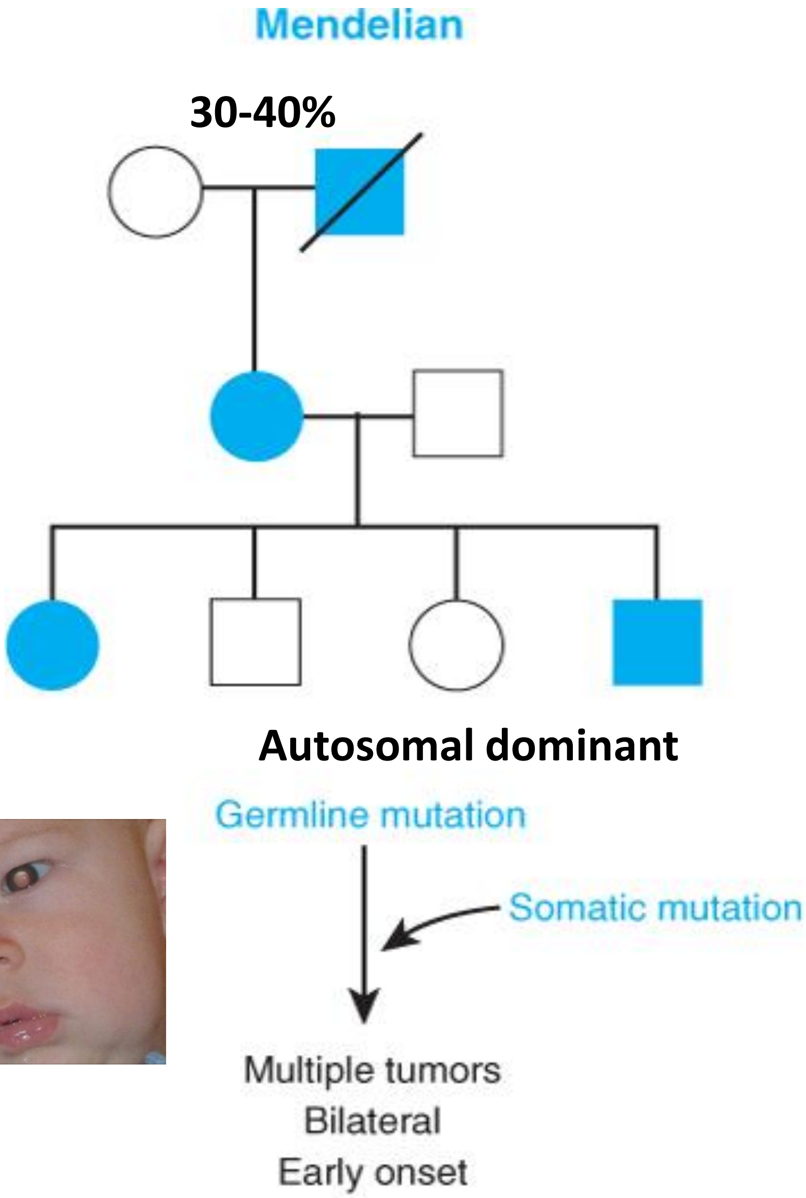
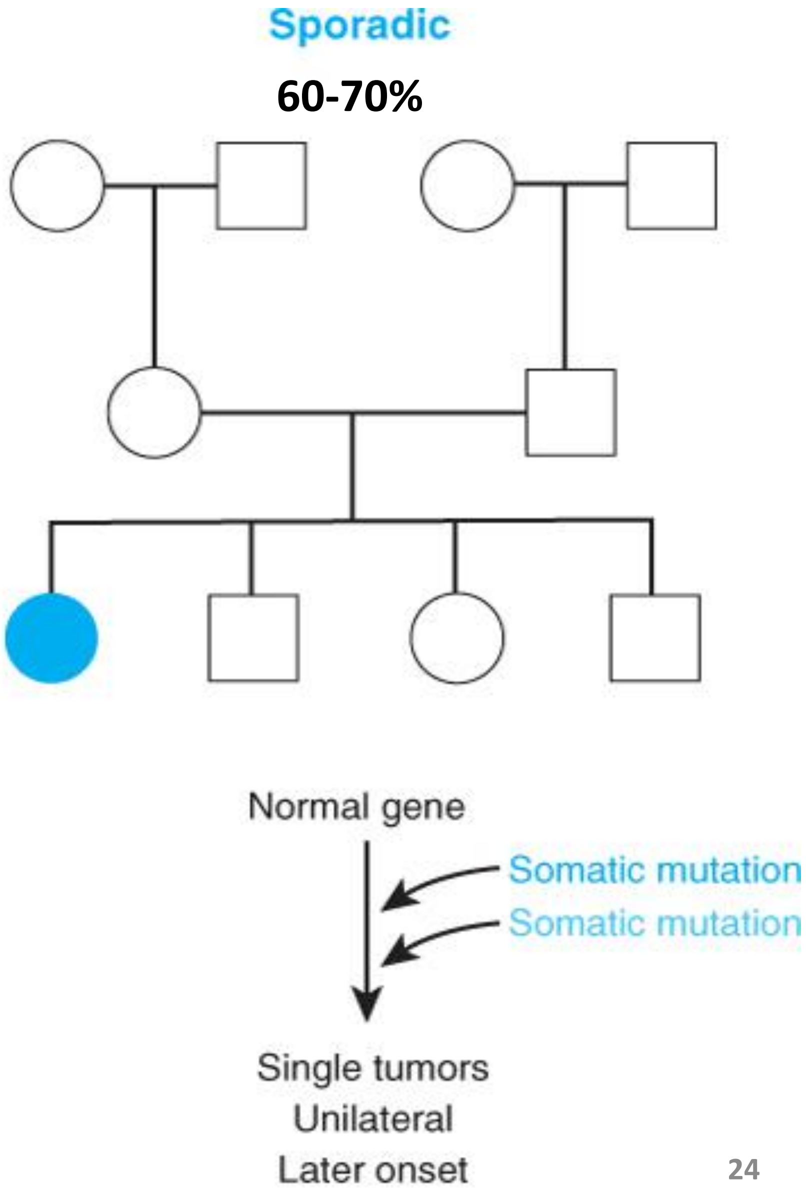
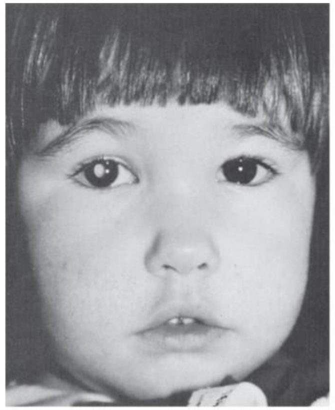
With familial cancers the first hit is inherited (usually germline mutation)

Figure 8.7 Two-hit model. A tumor ensues when two "hits" have occurred in the same cell lineage that knocks out both copies of a gene. Two rare events must occur to produce a sporadic tumor. A person who is born with the first mutation, however, needs to acquire only one additional mutation to develop a tumor.

Differences between sporadic and inherited retinoblastoma





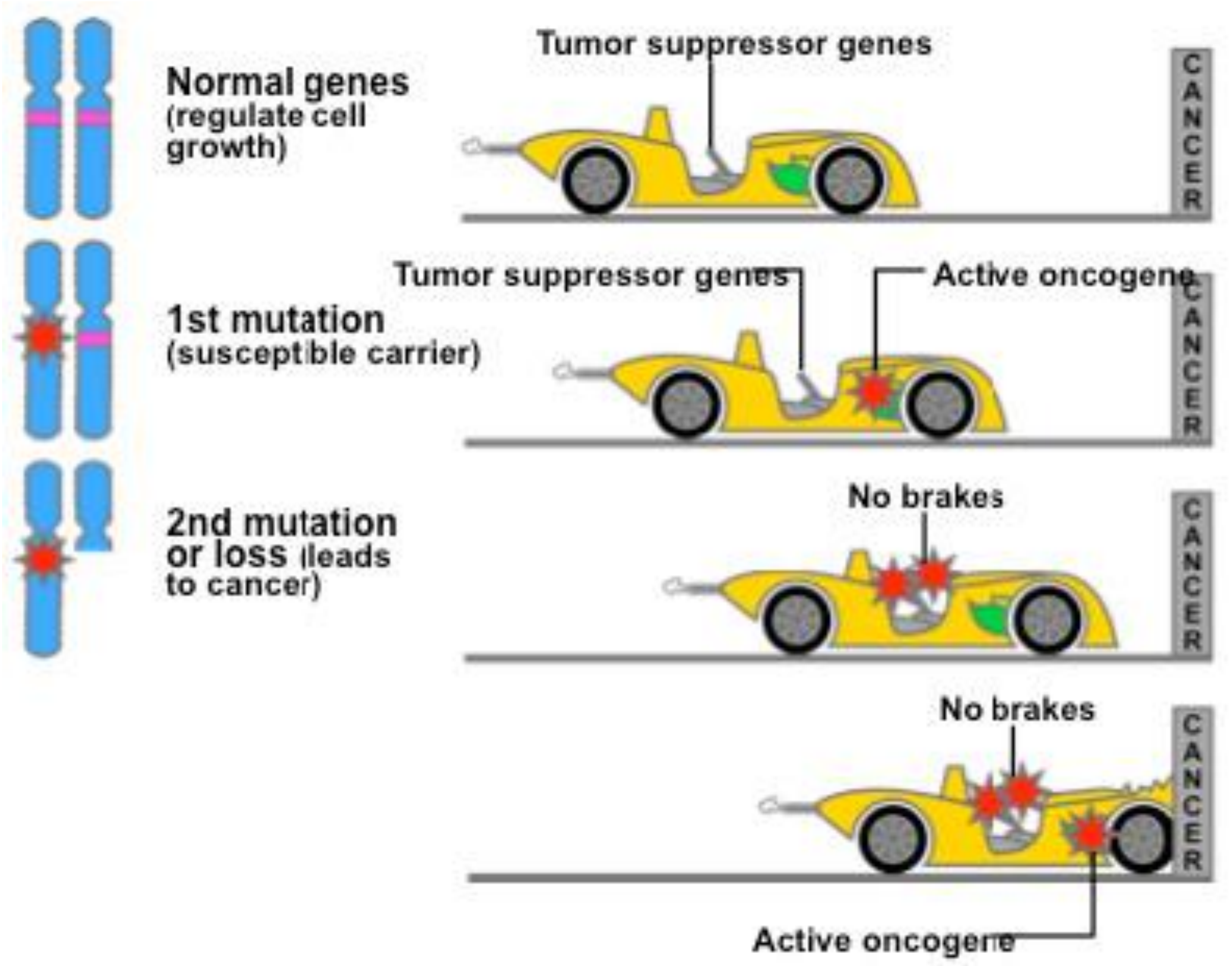


Differences between familial and non-familial (sporadic) cancer

Familial form vs sporadic form:

- Often multiple tumors and bilateral (familial)
- Familial form present at an earlier age than sporadic form
- Family history often present (one parent is usually affected in familial form)

Tumor suppressor mutation = “NO Brakes”



p53 is a *tumor suppressor protein*

p53

Transcription factor activated by cell stress, particularly DNA damage and it activates the G1/S checkpoint

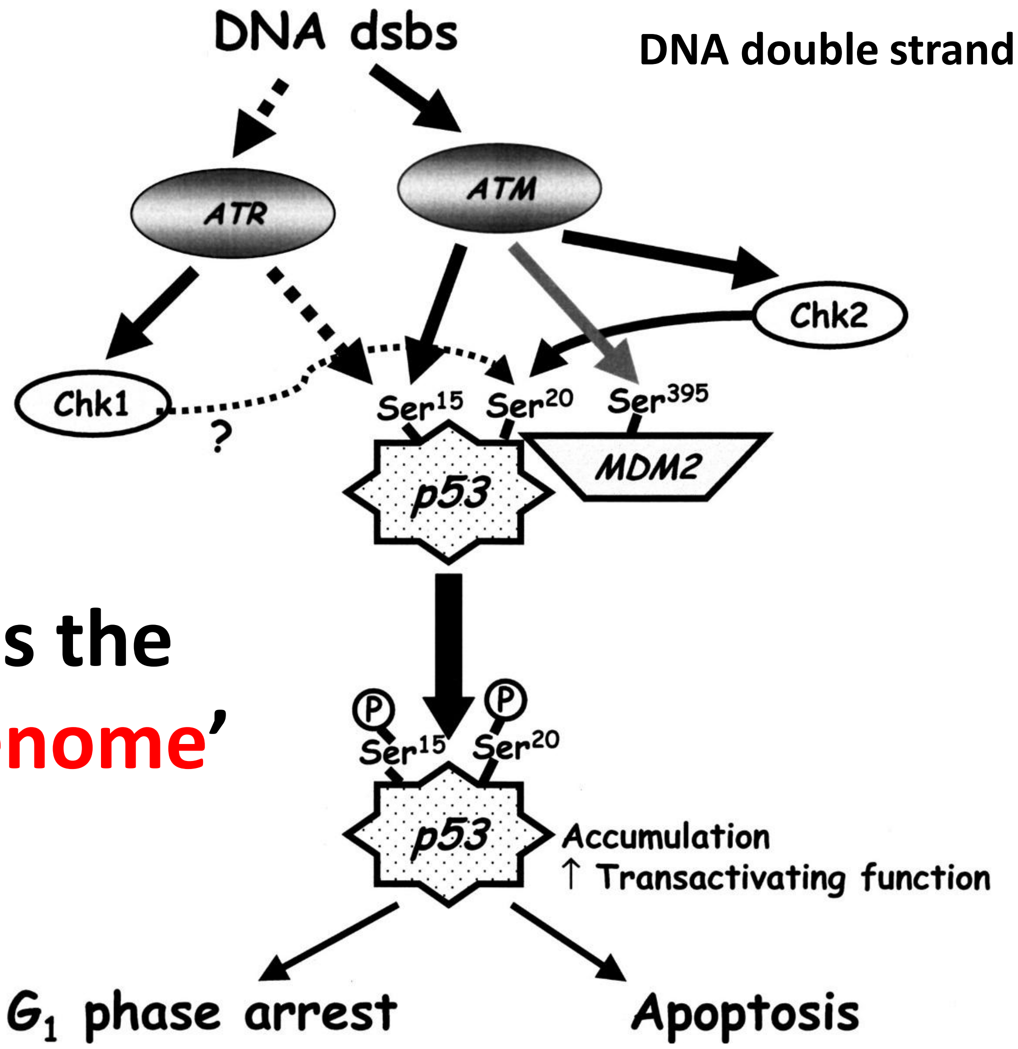
Cell Cycle Control
(Cell Birth)

Apoptosis Control
(Cell Death)

Loss of p53 function results in **net cell growth**, and **increased mutation** frequency, which drives cancer progression.

Mutated in more than 50% of all cancers.

Role of p53 in signaling through the G1 checkpoint



DNA double strand breaks

p53 is regarded as the 'guardian of the genome'

p53 transcriptional targets by class

Class of genes	Name of gene	Function of gene product
p53 antagonist	*MDM2/HDM2*	induces p53 ubiquitylation
Growth arrest genes	*p21^{Cip1}*	inhibitor of CDKs, DNA polymerase
	Siah-1	aids β-catenin degradation
	14-3-3σ	sequesters cyclin B–CDC2 in cytoplasm
	Reprimo	G_2 arrest
	p53R2	ribonucleotide reductase—biosynthesis of DNA precursors
	XPE/DDB2	global NER
	XPC	global NER
	XPG	global NER, TCR
	GADD45	global NER ?
	DNA pol κ	error-prone DNA polymerase
Regulators of apoptosis	*BAX*	mitochondrial pore protein
	PUMA	BH3-only mitochondrial pore protein
	NOXA	BH3-only mitochondrial pore protein
	p53AIP1	dissipates mitochondrial membrane potential
	Killer/DR5	cell surface death receptor
	PIDD	death domain protein
	PERP	pro-apoptotic transmembrane protein
	APAF1	activator of caspase-9
	NF-κB	transcription factor, mediator of TNF signaling
	Fas/APO1	death receptor
	PIG3	mitochondrial oxidation/reduction control
	PTEN	reduces levels of the anti-apoptotic PIP_3
	Bcl-2	(repression of) its expression
	IGF-1R	(repression of) its expression
	IGFBP-3	IGF-1–sequestering protein
Anti-angiogenic proteins	*TSP-1* (thrombospondin)	antagonist of angiogenesis

Slows cell cycle (G1 arrest), allows time to do repairs

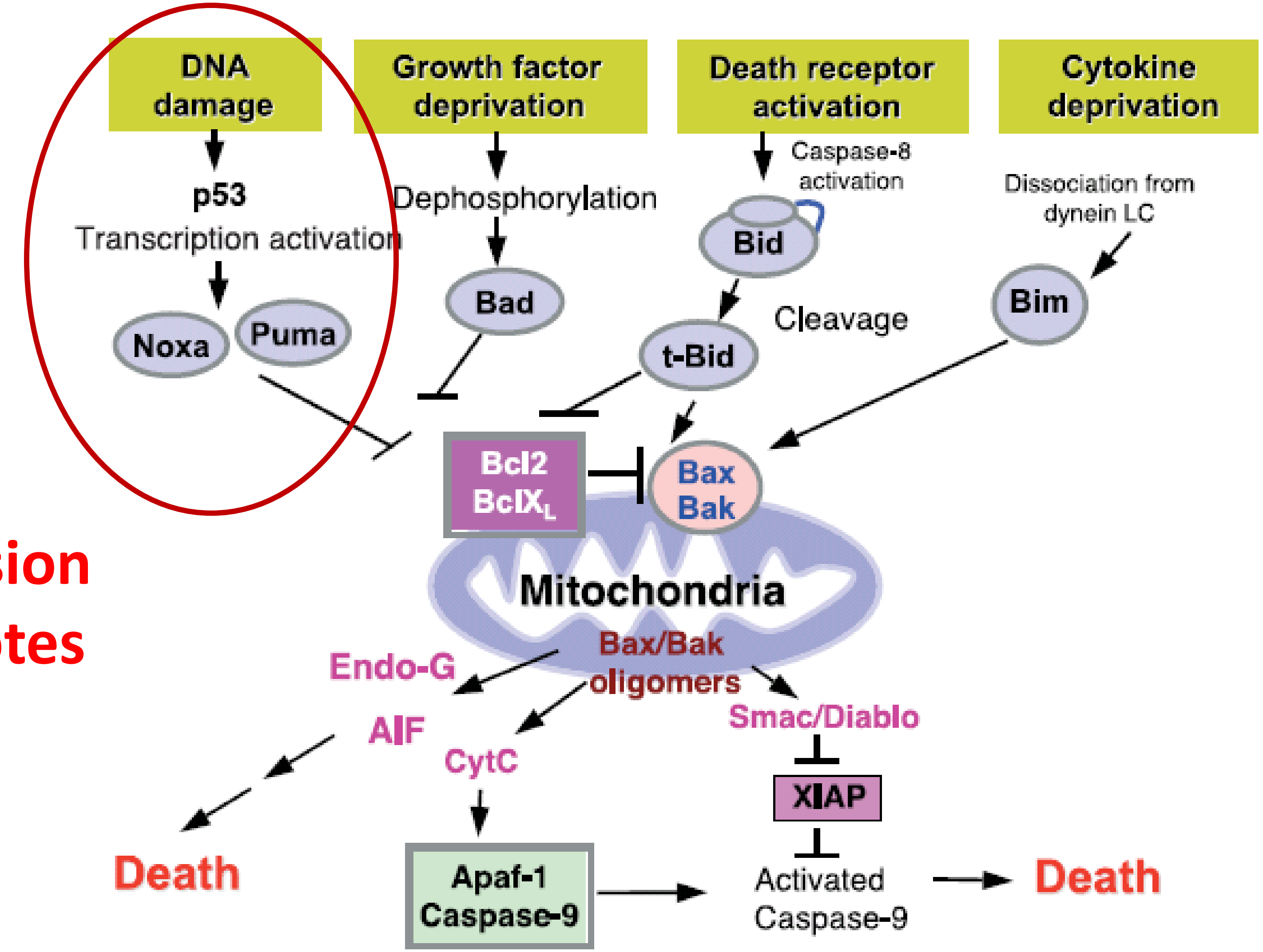
Increases DNA repair capabilities

If damage is too great then initiates apoptosis and eliminates cell

p53 has the 'power of life and death'

Table 9.2 *The Biology of Cancer* (© Garland Science 2007)

p53 activates the intrinsic apoptotic pathway



p53 **drives expression of Bax that promotes apoptosis**

p53 can influence apoptosis in several ways

p53 increases expression of:

- Bax (promotes apoptosis)
- Fas receptor (CD95)
- IGFBP-3 (sequesters cell survival proteins like IGF1/2 away from receptors)
- Activates intrinsic and extrinsic pathways for apoptosis
- (**More detail in DLA on Apoptosis in cancer**)

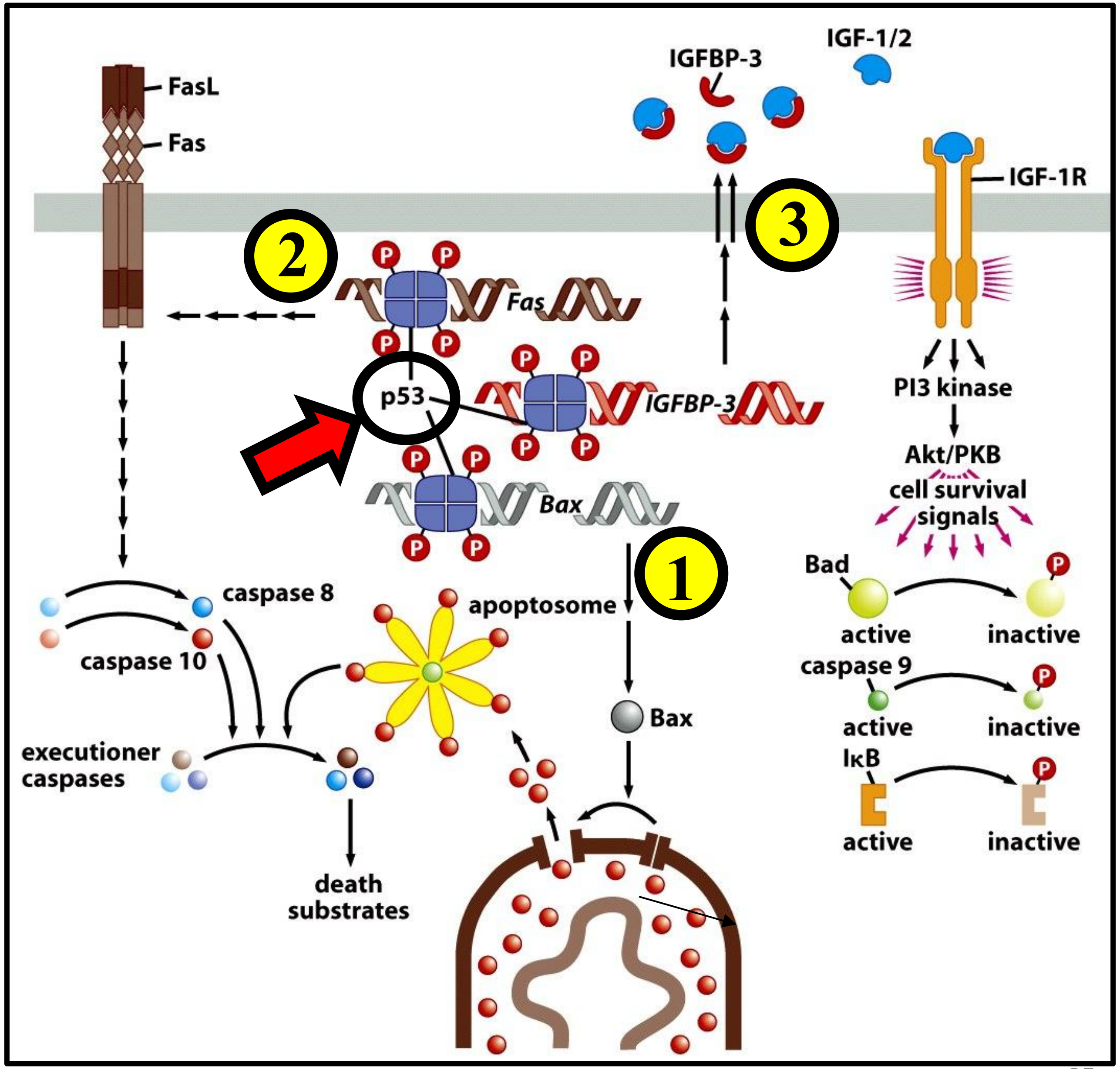
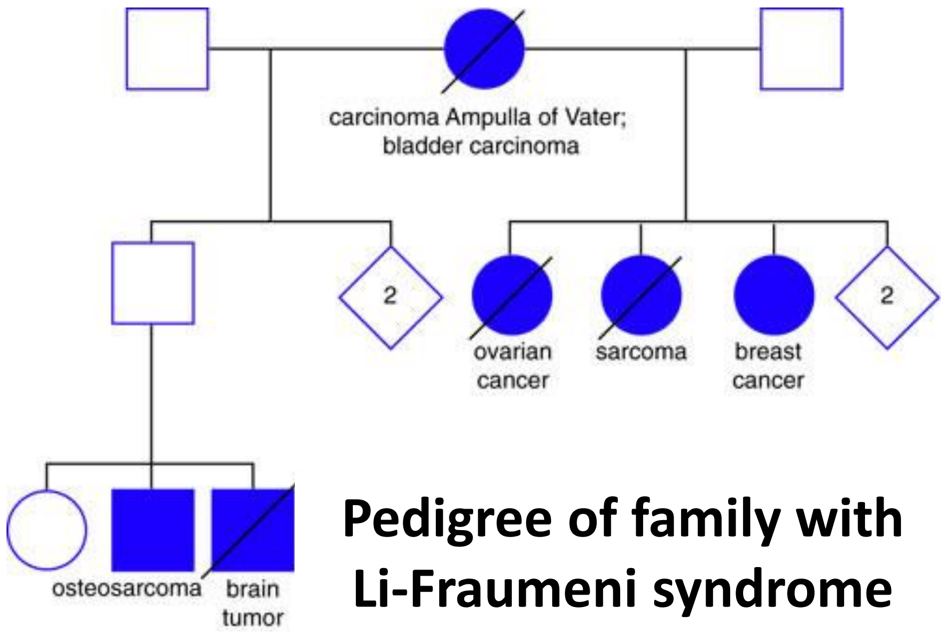


Figure 9.33 *The Biology of Cancer* (© Garland Science 2007)

Genetic basis of Li-Fraumeni syndrome: (Inherited mutation in p53)

- Rare disorder that increases the risk of cancer at a young age
- Results in several kinds of cancer including breast, bone, brain, and soft-tissue carcinomas
- 1st hit is inherited from Mom
- 2nd hit is somatic (responsible for LOH)



Pedigree of family with Li-Fraumeni syndrome

Families with Li-Fraumeni syndrome have inherited mutations in the TP53 gene

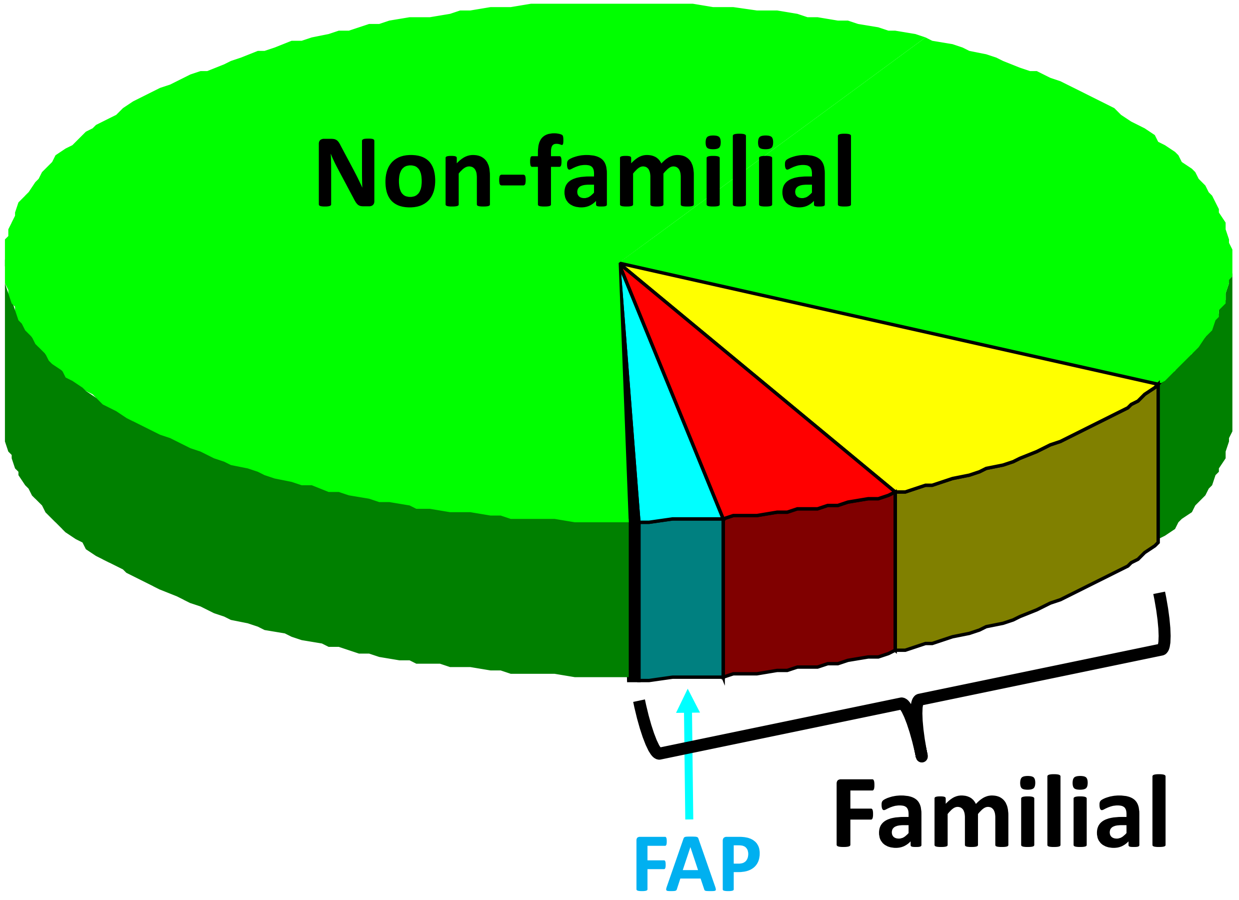
Common Inherited Cancers

Colorectal cancer

- Familial Adenomatous Polyposis (FAP)
- Hereditary Non-Polyposis Colon Cancer (HNPCC): Lynch syndrome

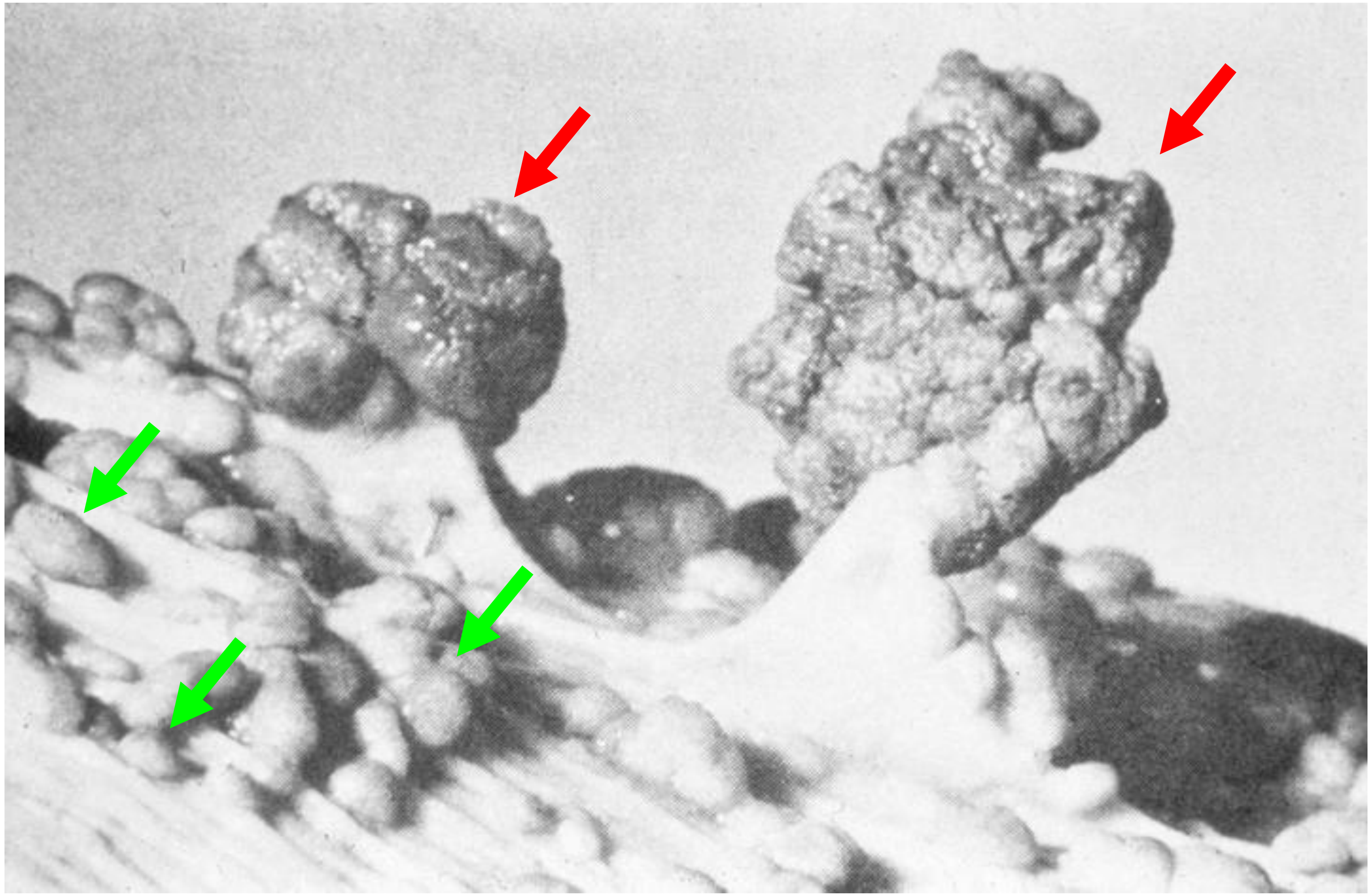
Breast/Ovarian

Familial and Non-familial Colorectal Cancers



Familial Adenomatous Polyposis

Colon from a patient with FAP

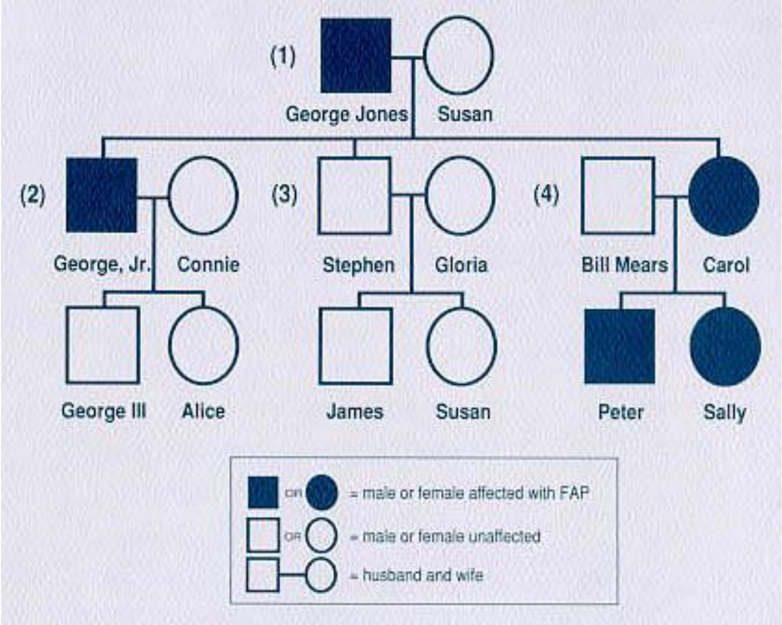


Numerous (>100) adenomatous polyps

Mutations in the APC (Adenomatous Polyposis Coli) gene on chromosome 5 (5q21)

Familial Adenomatous Polyposis

- Multiple (>100) adenomatous polyps develop throughout distal colon
- Very high penetrance
- ~1% of colorectal cancers
- Tumor suppressor gene (*affects* β-*catenin involved in growth control pathway*)
- Autosomal dominant inheritance

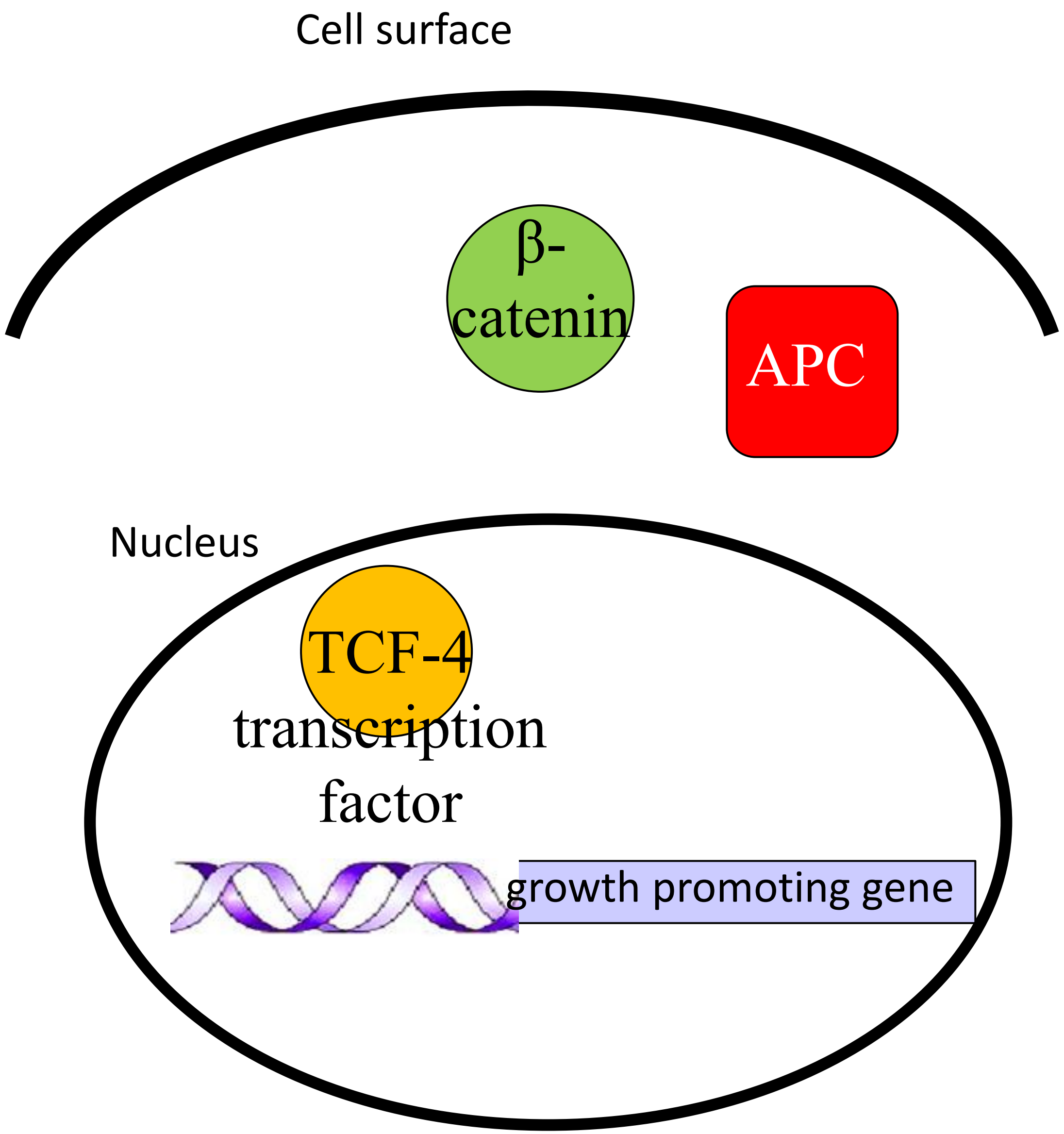


Pedigree of a family affected with Familial adenomatous polyposis

Many different mutations of the APC gene are reported (Allelic heterogeneity)

Normal APC Pathway

- APC a component of the WNT signaling pathway
- APC encodes a *tumor suppressor* whose role is to down-regulate growth promoting signals



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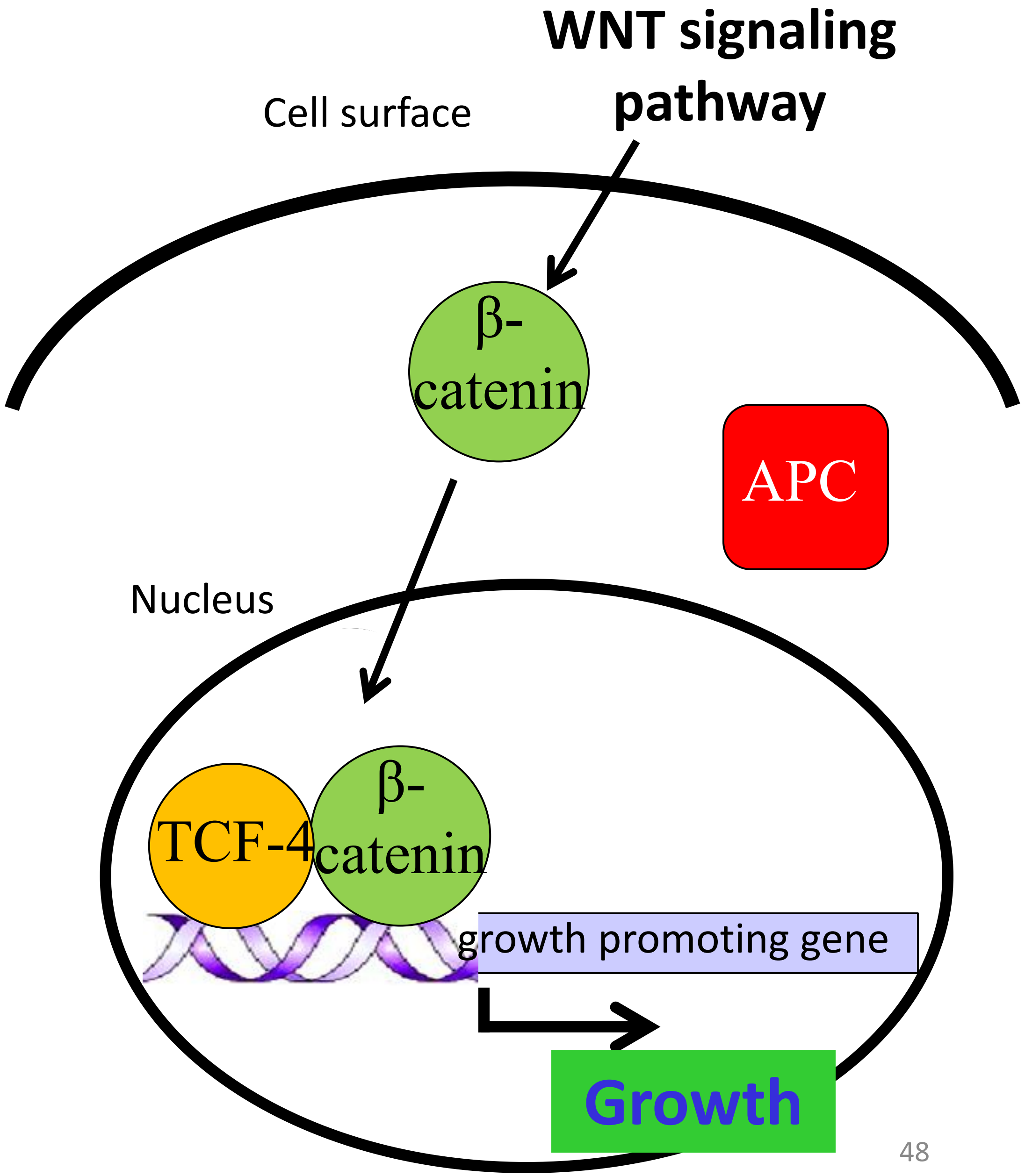
Normal APC Pathway: WNT signal present

- APC is a component of the WNT signaling pathway

When WNT signal present

- Destruction complex inactivated
- β-catenin not degraded
- β-catenin moves to nucleus, forms complex with TCF-4
- Activates growth promoting genes

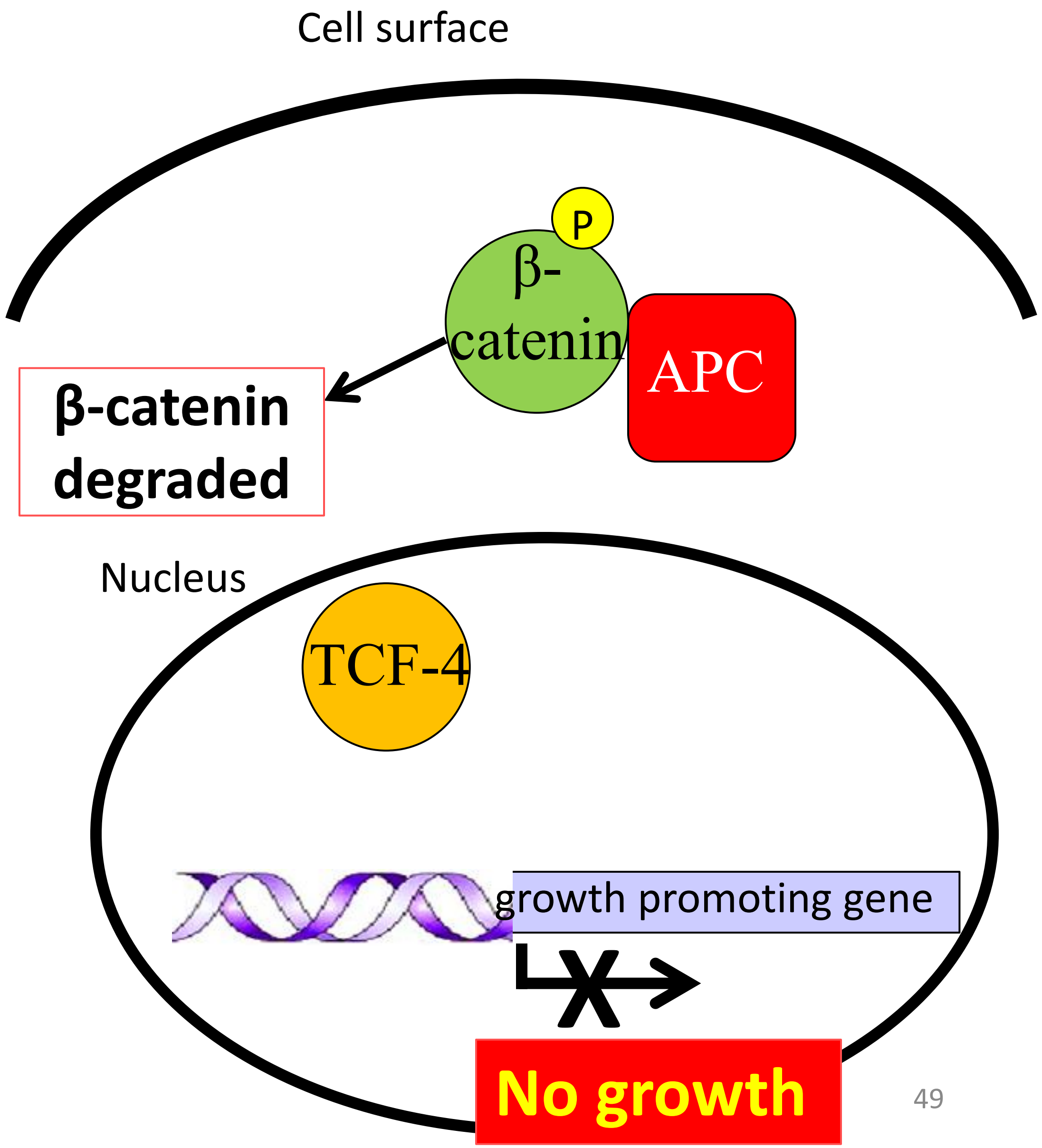


APC Pathway: No WNT signal

- APC is a component of the WNT signalling pathway

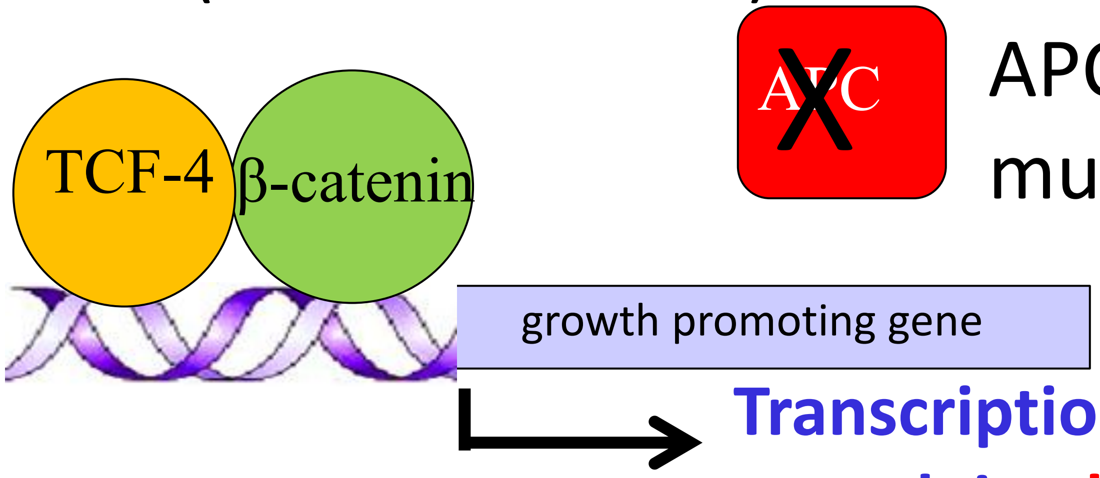
NO WNT signal

- APC interacts with β-catenin
- Triggers phosphorylation of β-catenin
- Ubiquitination and β-catenin degradation
- Resulting in low β-catenin levels
- APC acts like a **'gatekeeper'**



Disrupting the APC Pathway

(88% of mutations in FAP)

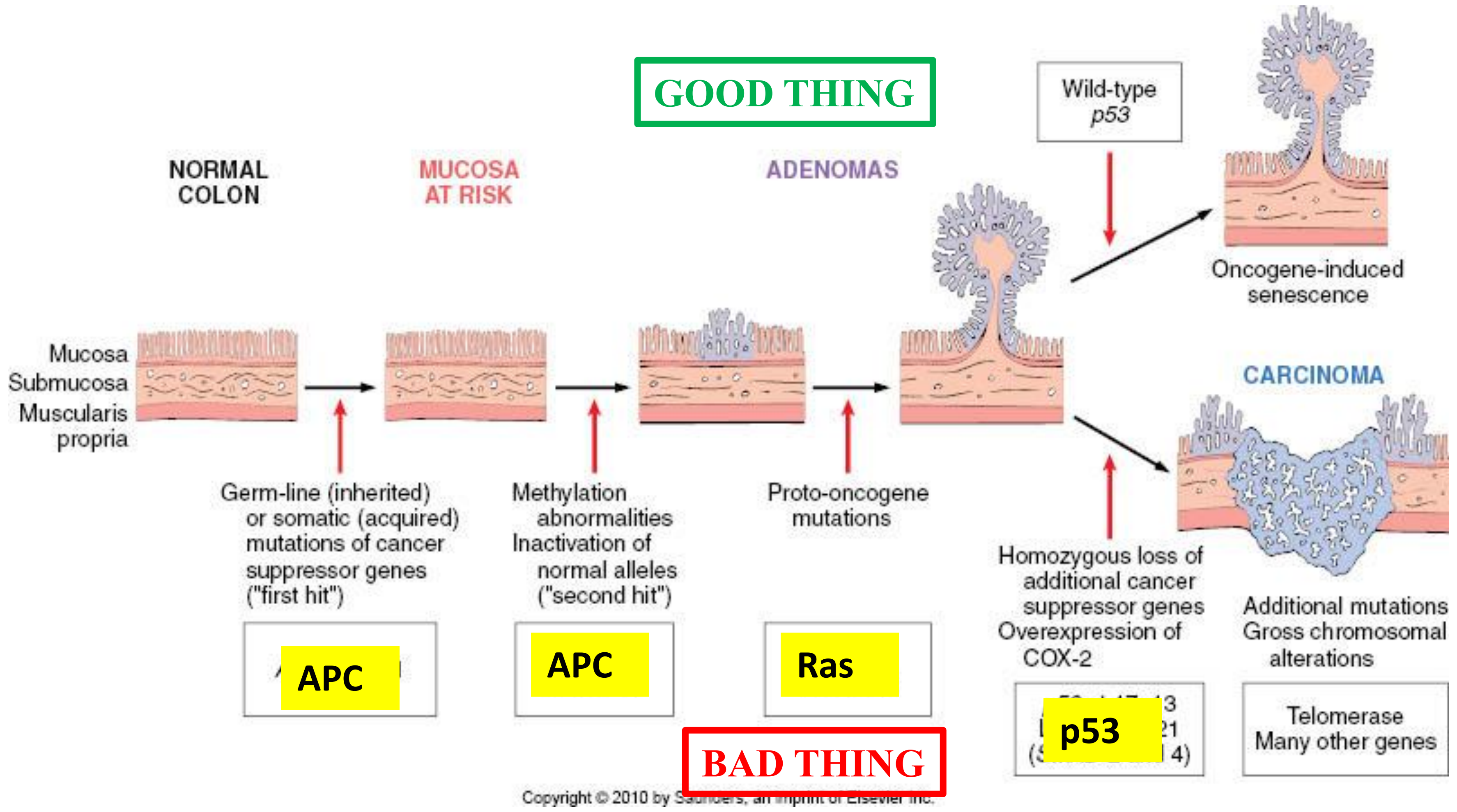


APC mutation

Transcription and growth in absence of WNT signal

- Mutant APC does NOT interact with β-catenin in absence of WNT signal (refer previous slide)
- β-catenin is NOT phosphorylated and NOT degraded
- β-catenin interacts with TCF-4 and activates **GROWTH** (in the absence of WNT signal)

Progressive/Cumulative Nature of Cancer in Familial Adenomatous Polyposis



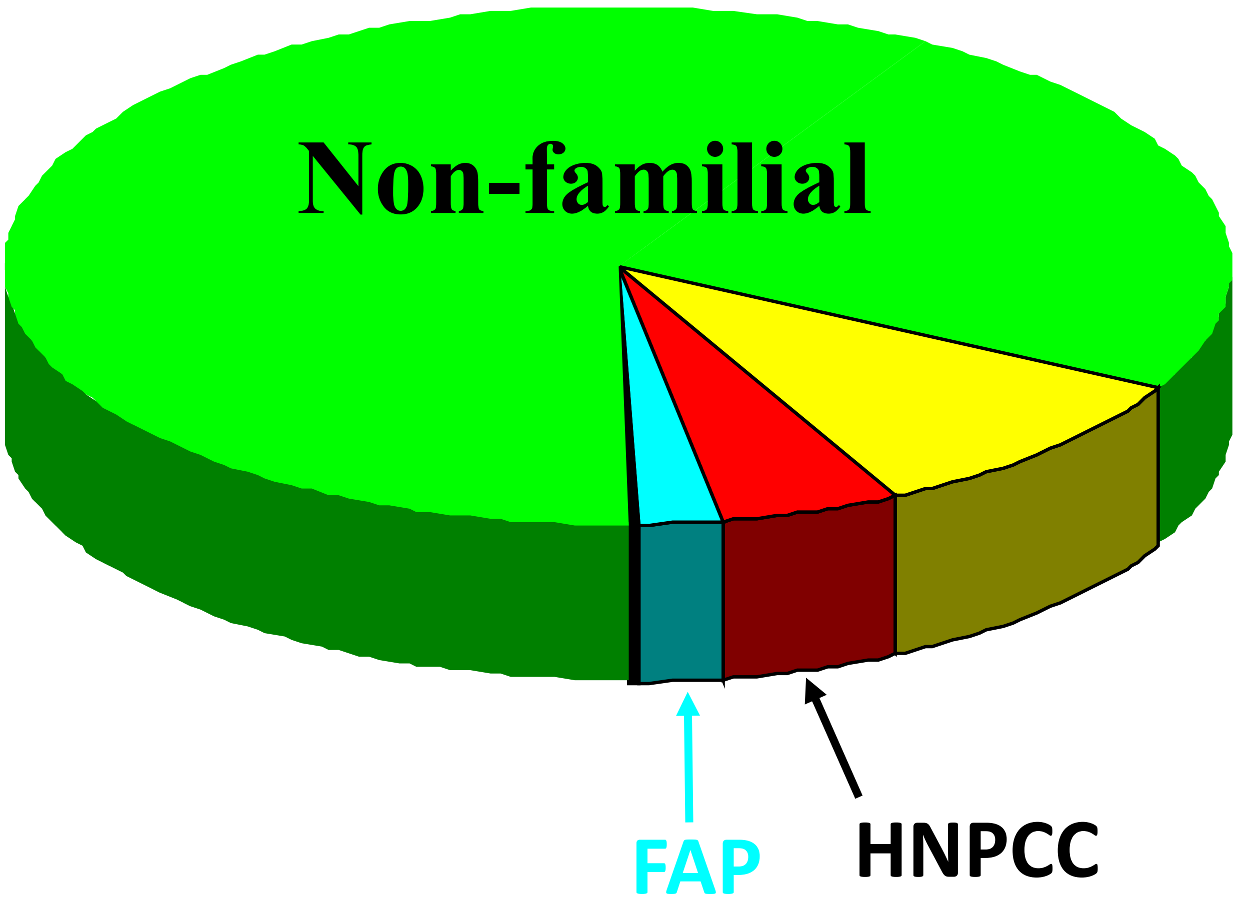
APC mutation is an early 'Gatekeeper' event. Loss of *p53* is a late event

Top right, cells that gain oncogene signaling without loss of p53 eventually enter oncogene-induced senescence.

Cancer Causing Genes

1. Oncogenes – normally stimulate growth
2. Tumor Suppressor genes – which normally inhibit growth
3. **DNA Repair genes - normally limit mutations**

Familial Colorectal Cancers



Hereditary Non-Polyposis Colon Cancer – defects in mismatch DNA repair

FAP vs. HNPCC

Many (hundreds) polyps, but progress slowly

Few polyps, but progress rapidly

Hereditary Non-Polyposis Colon Cancer (HNPCC): Lynch Syndrome

- Mutation of DNA mismatch repair (MMR) genes (Caretaker genes)
- At least five genes can be responsible (Locus heterogeneity)
  - MSH2 chromosome 2 ~60% of known mutations
  - MLH1 chromosome 3 ~30% of known mutations
  - Other MMR genes combined account for 5-8%
- Genes/proteins not directly involved in control of cell division **= mutator genes**
- Cells accumulate mutations at rates up to 1000 times higher than normal
- Tumors also exhibit **microsatellite instability.** (Short repetitive sequences of DNA)

FAP	HNPCC
APC mutation	MSH2/ MLH1 mutation
Accelerated	Normal
Numerous	Fewer polyps
Normal	Accelerated
	Microsatellite instability

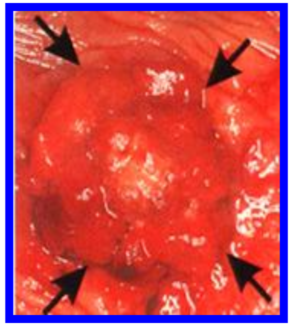
Normal Colon

Tumor Initiation ↓ *APC*

Polyp

Tumor Progression ↓ *RAS, p53,…*

Cancer



Gatekeeper vs Caretakers

Breast Cancer/Ovarian Cancer

- Incidence 1/12 in western society
- 1/3 affected women go on to develop aggressive tumors
- Up to 20% have familial history
- Lifetime penetrance about 85% for breast and 55% (BRCA1) or 25% (BRCA2) for ovarian cancer
- Pancreatic and other cancers also more common
- Male breast and prostate cancer incidence is also increased
- Two significant mutant genes identified so far
    - BRCA1 and BRCA2 (Locus heterogeneity)

BRCA1 and BRCA 2

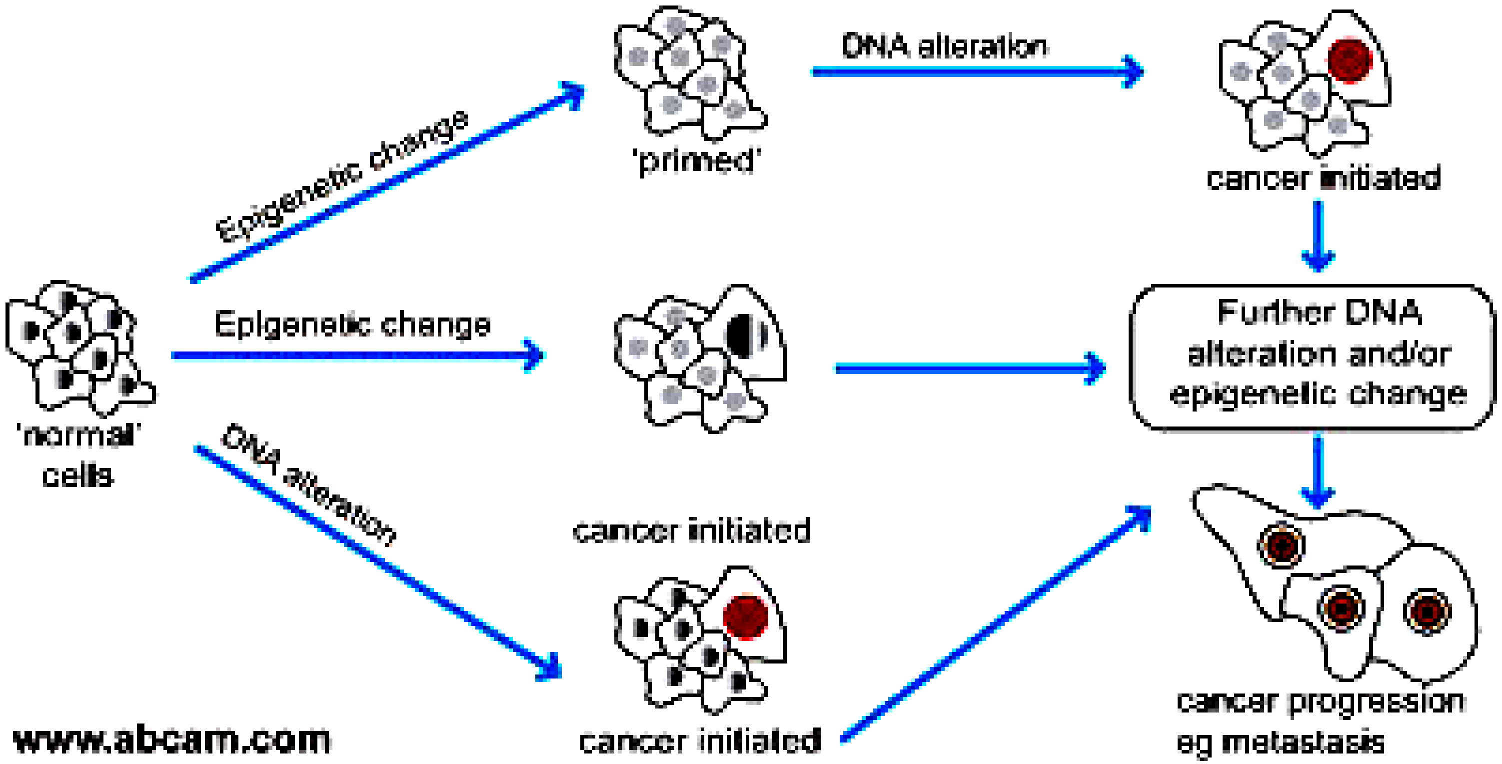
- **Genes code for breast cancer type 1 and 2 susceptibility protein**
  - Found in the cells of breast and other tissue
  - Involved in **DNA repair or apoptosis** when DNA can't be repaired
  - Loss of BRCA1/2, cells can duplicate with DNA damage and predisposes to cancer
  - Researchers have identified hundreds of mutations in the *BRCA1* gene, many of which are associated with an increased predisposition to cancer (Allelic heterogeneity)
- **'Angelina Jolie Effect': BRCA Testing Doubles**

Single gene mutations associated with inherited susceptibility to breast cancer

Gene	**Percent of hereditary breast cancer**	**Breast cancer by age 70 years**
BRCA1 (Breast and ovarian cancer)	52%	40-90%
BRCA2 (Breast and ovarian cancer)	32%	30-90%
p53 (Li Fraumeni syndrome)	3%	>90%

Epigenetics in Tumorigenesis

- DNA methylation and histone modifications are frequently altered in tumor cells
- Tumor suppressor loci are frequently hypermethylated in cancer cells (silencing of tumor suppressor genes)



Possible roles of epigenetics in cancer initiation or progression

Possible Direct Roles of Epigenetics in Tumorigenesis

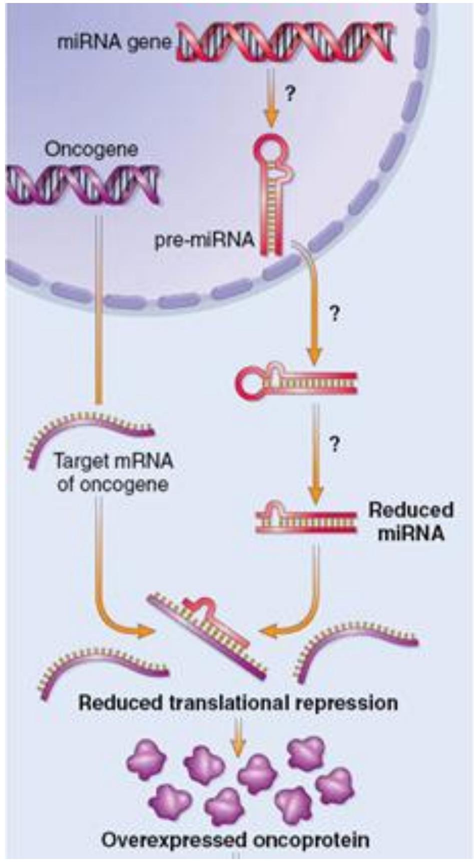
- Silencing of tumor suppressor loci causing cell overgrowth
- Loss of imprinting causing activation of growth associated genes or oncogenes
- MicroRNAs

MicroRNA in Tumorigenesis

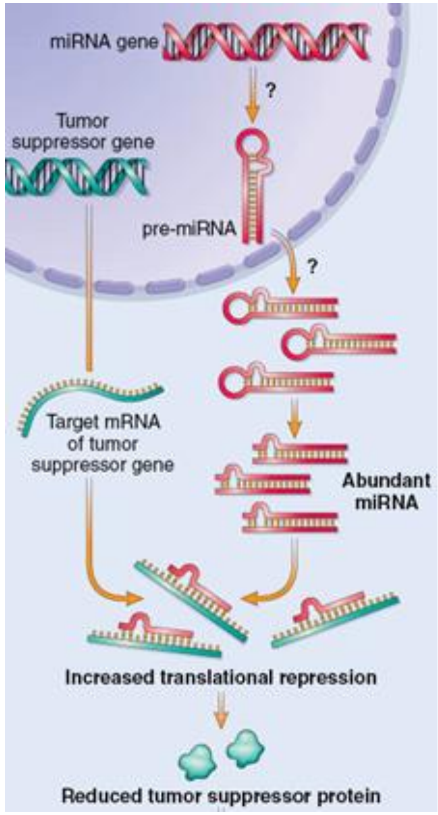
- miRNAs act to reduce the expression of genes by targeting specific mRNAs
- Aberrant expression/activity of miRNAs could be tumorigenic
- miRNAs have been shown to undergo changes in expression in cancer cells with frequent amplifications and deletions of miRNA loci

Reduction of miRNA that could have inhibited oncogene RNA

e.g. *miRNA-mediated upregulation of RAS in lung tumors and MYC in B-cell leukemias*



Increase in miRNA that inhibits tumor suppressor RNA



Expression array analysis and cancer

- Helps in cancer classification
- Useful to design appropriate therapy (HER2+)
- Requires molecular determination of genes being expressed in order to determine:
    - Rate of proliferation
    - Capacity for invasion
    - Potential of metastases
- Determining changes in the expression of large numbers of genes between two groups is possible with **Microarray Analysis**

Tumors may be grouped as:
-tumor vs. non-tumors
-benign vs. malignant
-primary vs. metastatic tumor

Thank you.

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